



TAI CHI
EXPLORE | RESEARCH | SUSTAIN



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TAI CHI NEWTECH INC AGROMACHINE DIVISION

“Integrated Efficient Solutions Provider”

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Land development - Farm Design - Construction - Pre Fab Housing

Poultry Farm Equipment And Project

Equipment Manufacturing - Installation - Commissioning - Operation Training



TAI CHI NEWTECH INCORPORATED COMPANY PROFILE

Tai Chi Newtech Incorporated (TCNI) is a world class multinational company that balances science of nature with cutting-edge product technology designed to provide value-added and research-driven feed ingredients. Along with its poultry farm and hatchery equipment, feed machine & parts, **TCNI** optimizes performance parameters in livestock and aquaculture. **TCNI** pioneers in the Philippines and has gradually expanded to neighboring Asian countries like Thailand, Vietnam, Indonesia, Malaysia and eventually in Europe to expand its global presence.

TCNI's strong brand value and trusted expertise is extended to livestock raisers and farmers. This is made possible through reliable professional teams and scientific partners dedicated to advancing animals' health, nutrition and production.

TCNI Agromachine Division is empowered by EI (Xingdao Xingyi Electronics Equipment Co., LTD.). EI is a subsidiary company of the 41st Institute of CETC which is affiliated with China Electronics Technology Group Corporation, one of the Fortune Global 500 companies. EI is committed to the research, production and service of incubation and hatchery equipment as well as poultry equipment innovation. EI equipment and projects are favored by its clientele in the industry. It is the choice brand for customers and is exported to 20 foreign markets, such as EU, America, Middle East, South East Asia, and Africa.

TCNI together along with EI will provide hatching equipment, poultry house automation equipment, other livestock climate control equipment, air filtration systems and other systems beneficial to the livestock industry. This partnership will provide an overall solution and turnkey projects for the livestock industry.

TCNI and EI will continue to provide an integrated and efficient solution with value added services and value-based products that will enhance the client's profitability and farm efficiency.

VISION



To be a leading and dependable organization contributing to food safety and healthy existence through innovation complemented by constant exploration and research for sustainable solutions.

MISSION



To provide value-added services and value-based products that shall enhance our client's profitability and farm efficiency.



To uphold and raise the bar of quality, safety, and pro-environmental products.



To endlessly pursue research innovations on feeds, feed technology, and farm efficiency concepts.



To forge bonds of trust and confidence with our customers while being mindful of their needs and satisfaction.

CORE VALUES

Tai Chi Newtech Inc. core values guide us on how we fulfill our mission. In the dynamic changes in the business field, our values remain constant. Our values inspire our personal and professional practices in our daily activities.



INTEGRITY

Absolute honesty, consistent and transparent professional work ethics will be our guiding force in our conduct towards our colleagues, clients, consumers and the public.



STRIVE FOR EXCELLENCE

Our continuous process of self-development is a personal responsibility we endeavor to achieve exemplary outcome and success.



POWER FOR INNOVATION

Our creativity, ingenuity, and exceptional desire to question common wisdom and to challenge the status quo fuel our commitment to provide practical, recognizable, and research driven solutions.



COMPANY INTRODUCTION

Qingdao Xingyi Electronic Equipment Co., Ltd. (www.eifhj.cn) is a subsidiary company of the 41st institute of CETC which affiliated to China Electronics Technology Group Corporation, one of Fortune Global 500 companies. Relying on deep military industry background, technological research and development, and support strength, we are committed to the research, production and service of incubation and breeding equipment. And also focus on design, matching and implementation of hatchery and breeding farms. After 3 decades development, the company has become the main supplier for breeding equipment and project.

Ever since launching the first EI incubator in 1984, through the joint efforts of all employees, the research field has been continuously expanded, the product system has been continuously improved, and products have covered most of the links in breeding industry chain. The company not only provides hatching equipment, automation equipment, poultry house climate control equipment, floor and cage breeding equipment, pig farm climate control equipment, air filtration system and project, but also provides customers with overall solutions and turnkey projects of modern hatchery and breeding farm.

EI series products have won the second prize of national "spark" and the United Nations invention and innovation prize, have over 100 national patents. EI brand has won the well-known trademark of China and the famous trademark of Shandong province. The company and its affiliated company, Bengbu EI Electronics Technology Co., Ltd. have been appraised by hi-tech enterprises for many years.

EI equipment and project are favored by the customers in the industry. And also have been exported to over 20 foreign markets, such as EU, America, Middle East, South East Asia and Africa. International influences, brand value and market share increasing year by year.

The company has R&D team both in Qingdao and Bengbu, totally more than 100 senior designers. The average age is less than 30 years old. They have foresight, advanced technology, advanced design means and strong technology accumulation, continuous innovation, independent research and development. With their efforts, advanced products and equipment are introduced to the market every year. Based on domestic marketing and vigorously expanding overseas markets, the company has built up a global sales service network with many offices at home and abroad. The company has a perfect after-sales service system, which can provide users with pre-sale consultation, advice and direction of hatching and breeding, technical training and after-sales service, and is committed to providing customers with more thoughtful service.

Creating value for customers is our management ideas and creating a century-old brand is our target. The company will continue to explore and study the system solution of intelligent breeding project for green and sustainable development, actively develop and promote environment-friendly intelligent breeding equipment and technology, faithfully fulfill social and industry responsibility, provide customers with super-value economic benefit, and contribute to the sustainable development of breeding and continuous environment improvement. Under the spirit of Solidarity, Integrity, Diligence and Innovation, the company will develop and progress together with our employees, customers and society.

CREATE VALUE FOR CUSTOMERS ESTABLISH CENTENARY BRAND



QINGDAO RESEARCH AND
PRODUCTON BASE



BENGBU RESEARCH AND
PRODUCTION BASE

CONTENT

COMPANY INTRODUCTION

PRODUCT CATALOGUE

Climate Control System of Poultry House

Broiler Cage Equipment

Floor Type Broiler Equipment

EI Cloud Service Platform

POULTRY FARM PROJECT

Steel Structure Poultry House

Feeding System

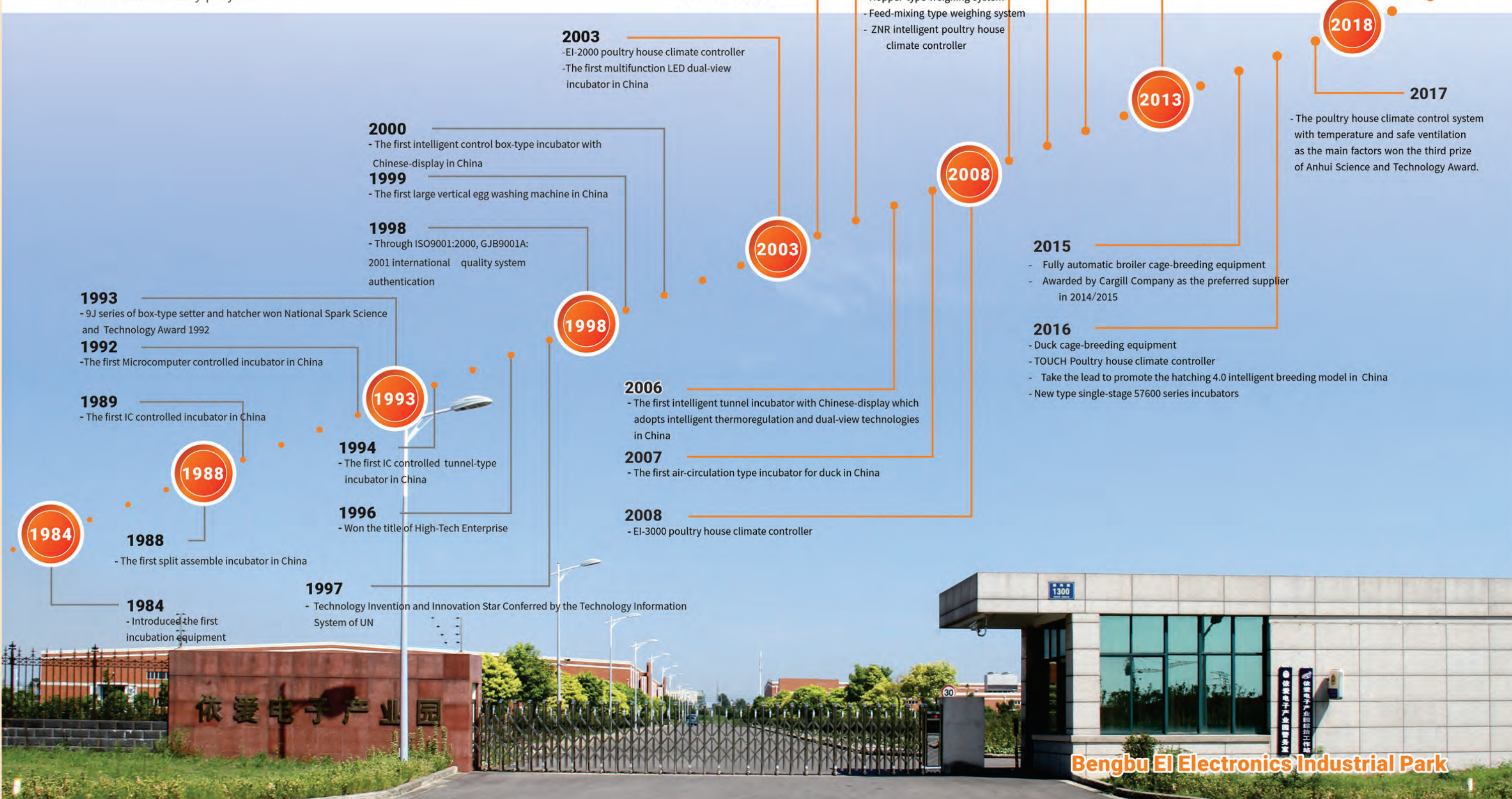
Heating System

Project Case



DEVELOPMENT HISTORY

Ever since launching the first incubator in 1984, we have been committed to R & D and production of incubation equipment, hatchery automation, hatchery climate control, poultry house climate control equipment, floor feeding and cage feeding systems and also focus on design and implementation of modern hatchery and poultry farms overall solutions and turnkey project.

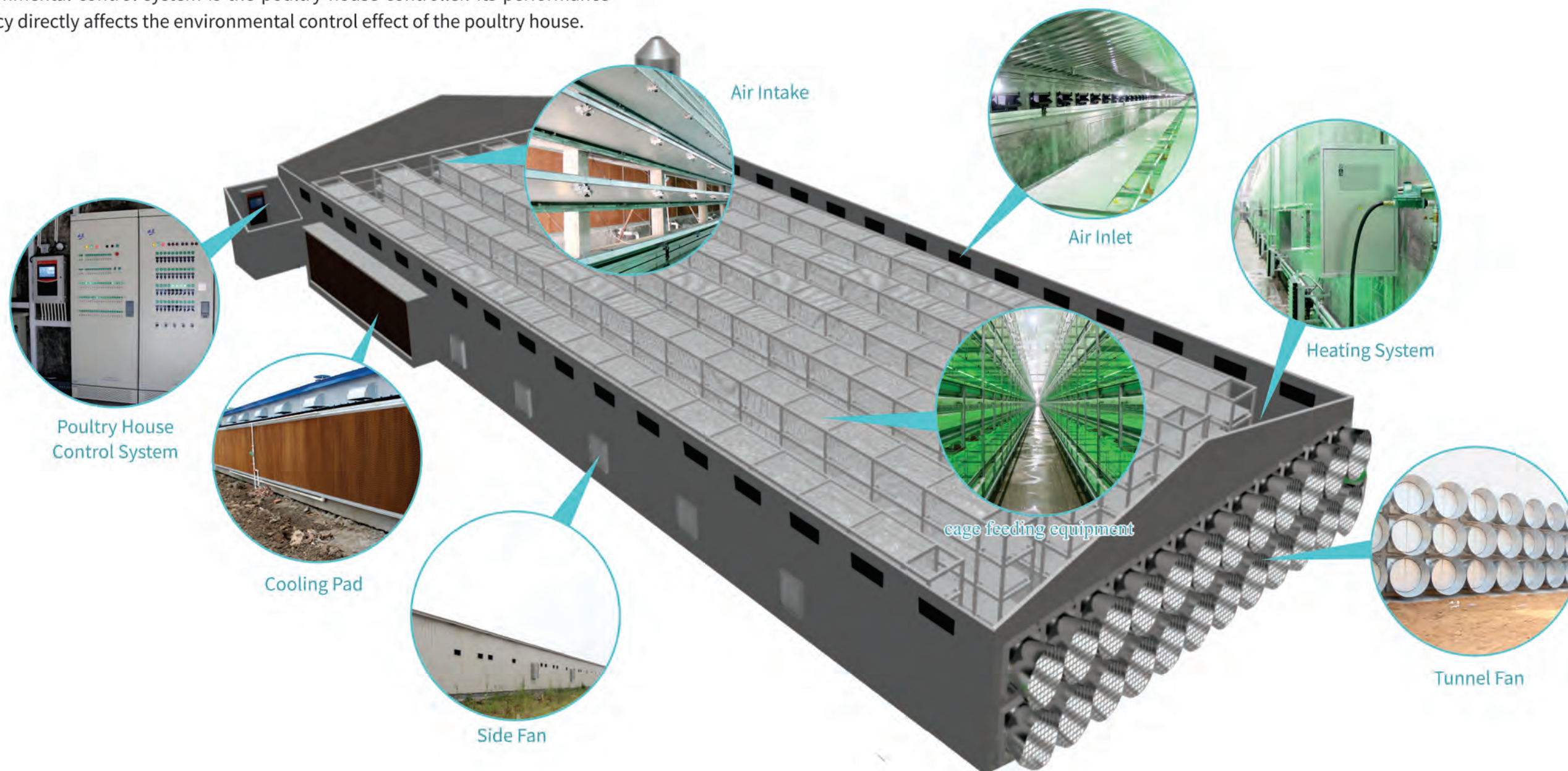




POULTRY HOUSE CLIMATE CONTROL SYSTEM

The poultry house climate control system controls the environment of the entire house by providing the proper cooling, heating, humidification, ventilation, negative pressure, lighting and other functions needed by the poultry house to provide sufficient fresh air and discharge the exhaust gas as well as the toxic gases. By setting the control parameters of the poultry house such as ventilation, temperature, humidity, static pressure and illumination, it can ensure sufficient oxygen in the poultry house, suitable temperature, moderate and reasonable illumination, it also meets the needs of poultry growth and maximizes the production of poultry. At the same time, it can reduce the artificial stress caused by artificial adjustments to the flock. The core control unit in the poultry house environmental control system is the poultry house controller. Its performance index and control accuracy directly affects the environmental control effect of the poultry house.

- ◆ POULTRY HOUSE CLIMATE CONTROLLER
- ◆ VENTILATION SYSTEM
- ◆ SPRAY HUMIDIFYING
- ◆ LIGHTING SYSTEM





POULTRY HOUSE CLIMATE CONTROL SYSTEM

Poultry house climate controller is a device that integrates data acquisition, function detection, intelligent calculation and control. The Controller collects temperature, humidity, carbon dioxide gas concentration, air pressure difference inside and outside the house, and the temperature outside the house. By combining the growth stage of the flock, the built-in intelligent algorithm coordinates the fans, air inlets, heaters, cooling pad and other equipment in an orderly fashion to provide the suitable temperature, humidity and light for the livestock. By meeting the demands of poultry growth, it effectively improves the production potential of flocks. The climate controller prevents the flock from stress caused by sudden environmental changes outside the poultry house.

The Climate controller provides the core decision making body for the essential components of the poultry house to provide for the optimal environment for the flock. With the use of the historical data provided and recorded by the Climate controller, farm management can be improved by correlating and comparing data

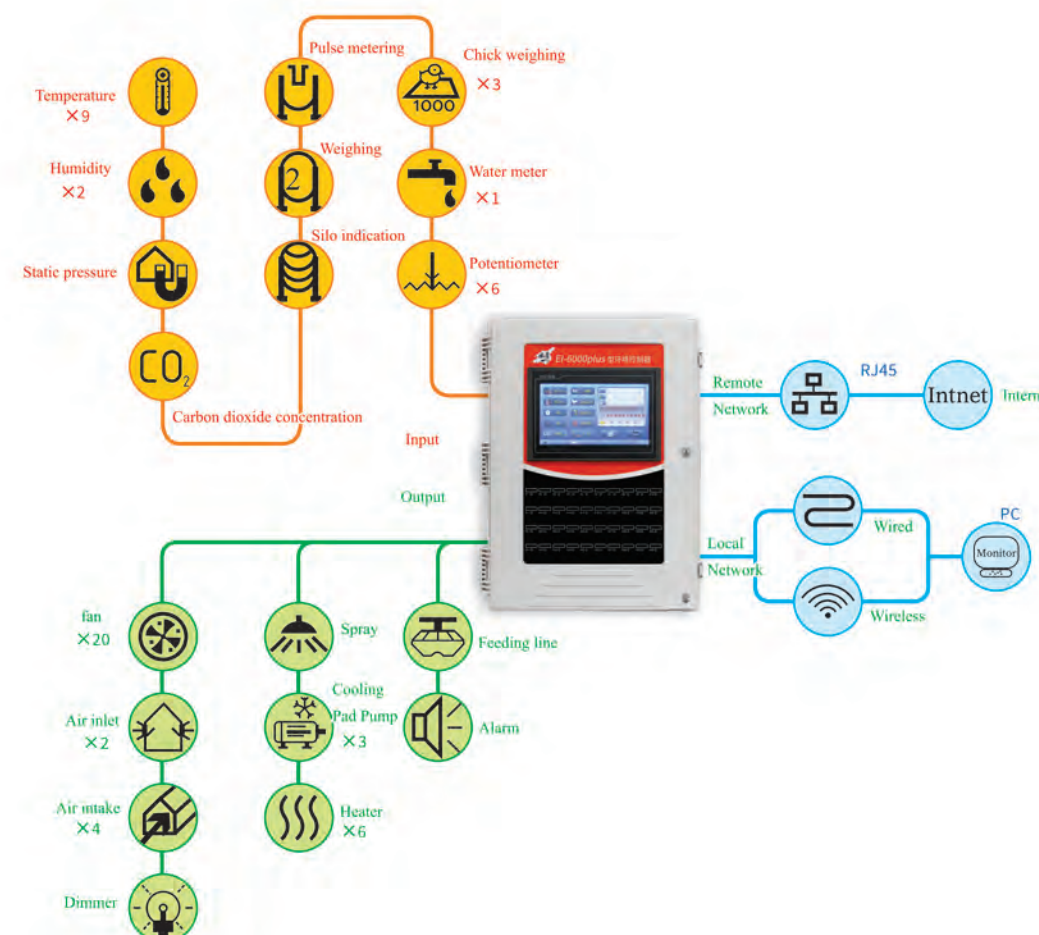
EI-6000 PLUS POULTRY HOUSE CLIMATE CONTROLLER

EI-6000 Plus is a new generation of controller with friendly and easy to operate interface. It has full-featured function and can run the poultry house in every aspect including temperature, humidity, CO2 concentration, static pressure and all kinds of ventilation modes and process/transition. It can record the supply and measurement of water and feeds, monitor poultry growth (weight) and mortality and other data supportive of farm management.



PRODUCT FEATURES

- With words, pictures, and curve. Touching type operation makes it intuitive, convenient and flexible.
- Limited to only two levels, similar to the Windows style which reduces users' learning curve
- 4 groups at most, and each group has 4 modules for users to select to adapt to different peripherals (heater).
- It's convenient for users to export or import data using USB drive.





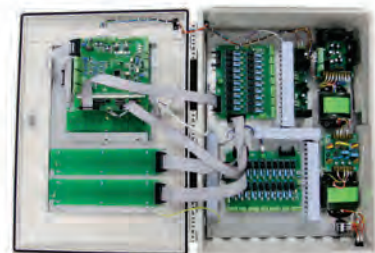
EI-6000 CLIMATE CONTROLLER



The EI-6000 climate controller is a new generation controller with multi-functions. It is practical, stable and highly reliable. A new temperature sensor has better long-term stability than the previous generation, and the calibration error is less than 0.5°C. Usage and maintenance is more convenient.

PRODUCT FEATURES

- The temperature sensor has good long-term stability. The calibration error is not more than 0.5°C.
- Comprehensive automatic control of equipment in the house (fan, light, feeding line, spray, heating, air inlet and wet cooling pad, etc.) to improve work efficiency and management.
- Automatic control of air inlet and cooling pad, can work either according to the static pressure setting or according to the ventilation table, or even a mixture of the two.
- The optional function is powerful, more temperature sensors can be selected to control different heating zones, and the air inlet opening degree can be automatically controlled by the optional static pressure module.
- The performance of the whole machine with good anti-interference and anti-lightning surge capability is stable and reliable. The test data of the electromagnetic compatibility test exceeds the national level 3 standard.



LIST OF CONTROLLER PARAMETERS

Index	EI-1000 Series	EI-6000	EI-6000 Plus
Temperature sensor	Maximum 3 indoor, 1 outdoor	Maximum 8 indoor, 1 outdoor	Maximum 8 indoor, 1 outdoor
Humidification	√	√	√
Cooling pad pump	1 channel	2 channels	3 channels
Heater zones	Maximum 3 zones	Maximum 6 zones	Maximum 6 zones
CO ₂	√	√	√
Negative pressure	√	√	√
Silo weighing	Not support	√(Silo weighing module required)	√(Silo weighing module required)
Chick weighing	Not support	Not support	√(Chick weighing module required)
Water & Feed metering	1 channel water metering	1 channel water metering, 1 channel feed metering	1 channel water metering, 1 channel feed metering
0-10V output	1000:Not support 1 channel for dimmer, 1 channel for FM-fan	Totally 2 channels, configurable, 1 dimmer, 1 FM-fan OR 2 both for FM-fan	Totally 2 channels, configurable, 1 dimmer, 1 FM-fan OR 2 both for FM-fan
Potentiometer	1 channel	2 channels	6 channels
Light	Timed on/off + dimmer	Timed on/off + dimmer	Timed on/off + dimmer
Ventilation level	Each level: 8 groups of fan, one channel of tunnel air intake, one channel of air inlets. Totally 10 levels.	Each level: 15 groups of fan, four channel of tunnel air intake, two channel of air inlets. Totally 20 levels.	Each level: 8 groups of fan, four channel of tunnel air intake, two channel of air inlets. Totally 30 levels.
Relay count	1000/1000B:16 1000C:20	32	41
Transmission module expansion	RS485	RS485	RS485
Parameter mutual copy	Group control	Group control	Group control, U disk
Historical data storage	Native + Group control computer+Cloud	Native + Group control computer now, Cloud's coming	Native + Group control computer now, Cloud's coming
Human-machine interaction	Keys+240*64 dots LCD	Keys+240*64 dots LCD	10 inch touch screen

APPLICATION SCENARIOS



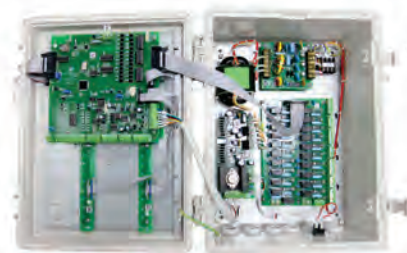
EI-1000 SERIES CLIMATE CONTROLLER



EI-1000 is a one of a kind, full-featured, economic, easy to operate, functional, practical, and robust poultry house climate controller. It meets the requirement of small and medium-size poultry farm managers/owners with specific requirements and expectations for controller functions.

PRODUCT FEATURES

- Through the automatic control of fan, lighting, feed line, spraying, heating and cooling pad it improves work and management efficiency.
- Aside from the automatic control air inlets and air intake, it also has the provides data on water usage, temperature record, network group control and alarm protection, etc.
- Additional functions are powerful. The speed of the variable fan can be controlled by the selection module and the air inlets and tunnel doors can be automatically controlled by the static pressure module.
- The Controller performance is stable and reliable, with good anti-interference and anti-lightning surge capability. Electromagnetic compatibility test data exceeds the national level 3 standard. Temperature sampling is stable and reliable and is capable of resisting the inverter interference and high frequency electromagnetic interference.





VENTILATION SYSTEM

50-INCH FAN



EI-F50 butterfly cone fan



EI-F50 shutter fan

50-inch Butterfly Cone Fan

The butterfly cone fan adopts the butterfly door opening and closing structure, this has minimum wind resistance, good sealing and high efficiency. Compared with the traditional shutter fan, the ventilation volume is increased by 15%, and the energy consumption is decreased by 12-15%.

50-inch galvanized shutter fan

Easy installation. When the fan starts, the opening mechanism opens the shutters automatically, and the angle level is opened with minimum wind resistance. The whole machine runs smoothly, has low vibration and noise while providing stable air volume. When the fan stops running, the shutters are closed and sealed by gravity and springs. The blades are evenly balanced and the shutters effectively seals the fan from external debris/insects.



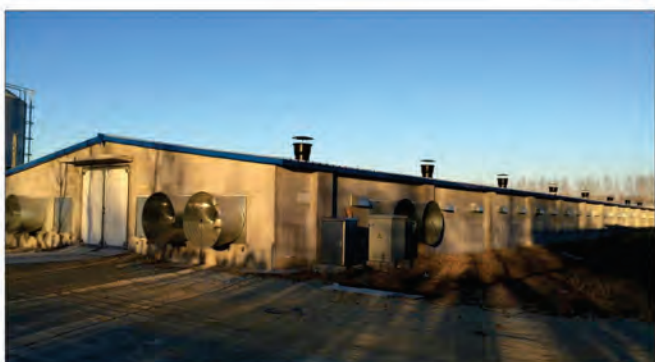
ROOF FAN



Roof fans are often used in poultry houses designed with vertical ventilation. This design type provides for a very low cross infection rate. They also have the functions of automatic frequency conversion and automatic opening of dampers. This is suitable for cold areas needing good ventilation uniformity. Complemented with air inlets, it ensures the most effective ventilation for poultry in winter.

PARAMETERS

- Ventilation rate: - 20Pa, 11800m³/h;
- Noise: ≤70 dB (within 1 m distance);
- Frequency conversion range: 0-50Hz;
- Waterproof grade: Ip55;
- Power: 750W;
- Working power: three-phase four-wire system, 380 VAC+10%, 50/60Hz .



COOLING PAD

The cooling pad is used in conjunction with the air inlets and the fans for evaporative cooling. Heat is easily absorbed after passing through the cooling pads. Temperature is decreased as air passes through the cooling pad, thereby reducing the temperature inside the house.



Cooling Pad

AIR INTAKE

Types of air intake: cord type, heat preservation door & gear rack drive type, heat preservation door & wire rope drive type.



Cord Type Air Intake

PRODUCT FEATURES

- The air intake installed in the ventilation outlet of the air room can effectively prevent the cold air from entering the chicken house during cold season. It can effectively control the wind speed by changing the opening of the air intake during warm/hot season.
- The inner core board of rack and wire rope drive type is made of polystyrene foam sandwich board. This provides very good thermal insulation and effectively regulates the internal temperature of the chicken house. The wall-mounted air intake can be designed with a single opening or double opening of upper and lower layers.
- Rack and wire rope drive type have good consistency of opening and closing. This equipment ensures good sealing with no air leakage and proper ventilation during various stages of bird growth.



Gear Rack Drive Type Air Intake



Cooling Pad Type Air Intake



AIR INLET

Air inlets provide fresh air to the poultry house and regulate the negative pressure inside the house. When fans are on, fresh air enters the poultry house through air inlets because of negative pressure. As the system turns on more fans, negative pressure increases, and this is tempered with the commensurate opening of inlets.



Air Inlet

PRODUCT FEATURES

- Dimension: 550*260*139mm
604*240*139mm
(mounting dimension 610mm*250mm)
- Material: Cover: ABS/PP
- Window: raw material contains anti-UV and anti-aging elements, long service life.
- Protection net: made of good-quality engineering plastic material, excellent performance, solid and durable.
- Automatic control and manual control for your option.

SPRAY HUMIDIFYING

The multistage filter device removes impurities and harmful elements in the water. The copper nozzle spreads evenly and rapidly into the air. The particle diameter of the mist is 10-40μm and the atomizing degree is 1-2.5 M. It has achieved the best atomization effect and has the functions of cooling, disinfecting, increasing humidity and so on.



Spray humidifier

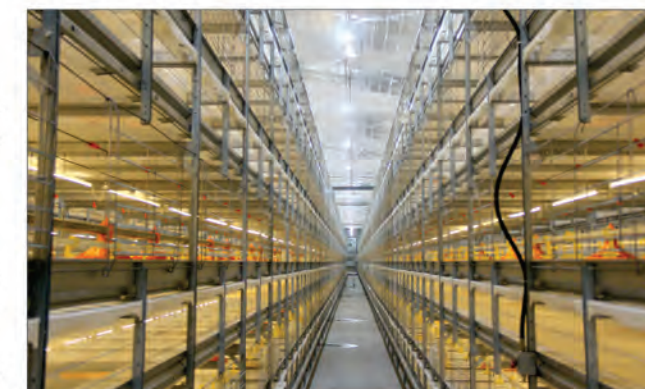


Spray effect

LIGHTING SYSTEM

The special LED lamp for poultry house not only has the advantages of energy-saving and long life of traditional LED lamp, but also has:

- Environmental applicability: It can adapt to the harsh environment of high temperature, high humidity, corrosive gas and high pressure washing in poultry houses.
- Structural diversification: It meets feeding requirements of different poultry houses.
- T8.0.6/1.2m fluorescent lamp with diffuse reflection and explosion-proof function, dimmable, flicker free, uniform illumination and suitable for floor breeding, the color temperature is 3000K and luminous efficiency>120lm/W.
- T5-1.2m fluorescent lamp with waterproof and fast connection function, dimmable, flicker free (thermal noise frequency>300Hz), uniform illumination and optimum Design Products of Serial Special optical Wavelength and Lamp Shading Angle, which is suitable for cascade cage breeding.
- E27 special breeding bulb lamp has special spectrum lines (such as warm white, white, red, white-blue switching and other special light), and the light source is easy to arrange and maintain. It is suitable for floor breeding and cascade cage breeding with high and low lights distributions in aisle.
- Special spectrum lines, LED special lighting spectrum is different from traditional lighting. Different light source products are used for different feeding targets, such as white, warm white, green, blue-green switching and other special light wavelength products.



T5 Warm White LED Lighting in Cage Breeding



T5 Green LED Lighting in Cage Breeding



T5 Warm LED Aisle Cross-shoot lighting in Cage Breeding



T8 Warm White LED Lighting in Floor Breeding



CAGE

Cage system is 2-4 times higher than the density of floor feeding system, so utilization rate of poultry house is significantly improved and land resources are saved. Meanwhile, feeding, feed supply, drinking, manure removal and birds harvest are automatically controlled.

The highly automated modern large-scale poultry farm consists of cages, automatic chick harvest, automatic manure removal, climate control, ventilation, LED lighting, heating, humidification, and centralized feed supplying, etc. The chick can be separate from manure with automatic manure removal and chick harvest system, so the breeding conditions are improved and drug dosage and drug residue are effectively reduced; All in all, cages can reduce the labor intensity, save production cost, bring significant economic benefits to poultry farm; reduce environmental pollution, create huge social benefits, and meet the demand of 'large-scale, standardized, industrialized and intensified' development.

◆ BROILER CAGE

FULLY AUTOMATIC BROILER CAGE

ECONOMY BROILER CAGE

AIR PASSAGE TYPE BROILER CAGE

ONE SIDE FEEDING BROILER CAGE

◆ DUCK CAGE

1 M DUCK CAGE

2 M DUCK CAGE

◆ LAYER CAGE

TRAVELING HOPPER CAGE

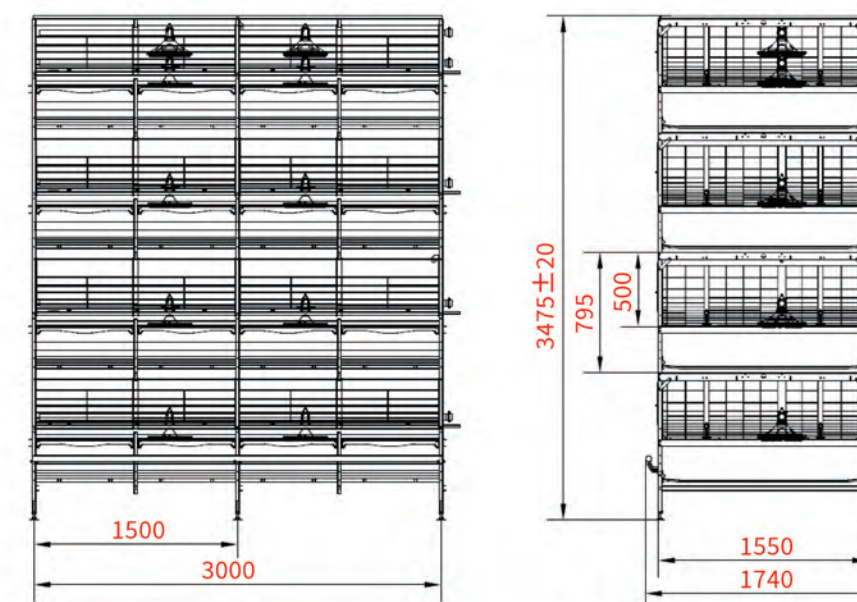
CHAIN FEEDING CAGE



BROILER CAGE

EI-LY/R15 TYPE FULL-AUTOMATIC BROILER CAGE

It is a full-automatic cage system for broiler rearing, which is composed of cage system, feeding system, manure removal system, poultry harvest system and so on. The automatic control of feeding, drinking, manure removal, bird harvest and climate for large-scale broiler feeding has been realized, which greatly saves labor and time.



PARAMETERS

- Unit length:3000mm
- Cage width:1550mm
- Cell height: 500mm
- Single floor height:795mm
- Three floors height:2660±20mm
- Four floors height: 3475±20mm
- Unit cage dimension(LxWxH):
3000mm×1550mm×500mm
- Unit cage drinker quantity: 12 birds
- Unit cage feed pan quantity: 2 birds
- Door Dimension: 720mm×330mm
(horizontal type)
- Unit cage breeding capacity: 93 birds
(2.5kg/bird)
- 3 tier breeding capacity:93×3=279 birds
(2.5kg/bird)
- 4 tier breeding capacity:93×4=372 birds
(2.5kg/bird)
- Automatic chick harvest efficiency:≥
6000 birds/h



PRODUCT ADVANTAGE HIGH QUALITY MATERIAL

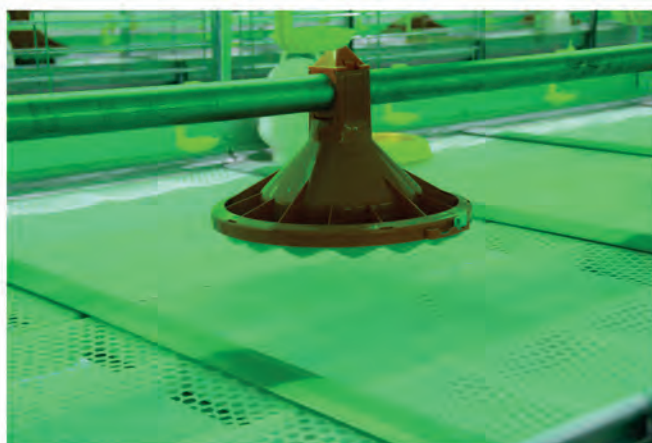
Chick cage structure parts are all processed by hot-dip galvanized steel plate, cage mesh is welded by zinc-aluminum alloy steel wire, anti-corrosion and durable with long service life





PLASTIC SLAT

The plastic floor has regular hexagonal mesh and is designed with patented technology making it soft and flexible. It is suitable for chicks to trample and leak manure. It can effectively protect chicks' chest and foot from injuries, and greatly improve the environmental of chick feet.



BROILER FEED PAN FOR CAGE

The pan is specially designed for full automatic broiler cage, with the feature of low height, simple disassembly, easy cleaning. It is suitable for feeding one-day old chick up to adult chick. Made of PP and food grade color master batch, 15 grills, and feed storage 1~1.5kg.



AUTOMATIC MANURE REMOVAL SYSTEM

The head and tail of the cage are equipped with a manure belt anti off-track device. Manure removal system has two scrapers. The tail end scraper adopts high wear-resistant polyurethane scraper. It can cleanly scrape manure without damaging the manure belt. The manure removal system is designed with seamless connection circular rubber conveyor belt, with high strength and long service life. The modular design of the roller rack is suitable for the width requirements of cage houses with various specifications.



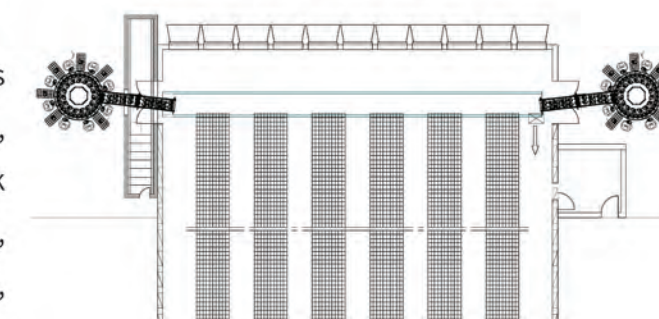
Manure belt and removal mechanism



Manure discharge system

AUTOMATIC CHICK HARVEST SYSTEM

The automatic chick harvest system is composed of cage manure removal system, cross chick delivery device, oblique chick delivery device and chick catch turntable, conveying over 6,000 birds per hour. In addition, the two-way chick delivery system can be configured to increase efficiency to 10,000 birds per hour. It is more flexible and more efficient, allowing independent delivery of different tiers in different lines, which cannot be achieved in one-way system.



Two-way chick harvest diagram



Chick harvest by manure belt



Turntable catch

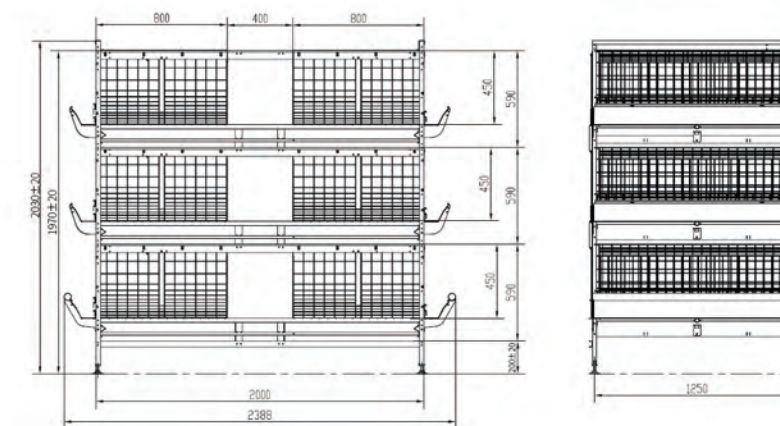


EI-LY/F7040/8040 AIR PASSAGE TYPE CAGE

EI-LY/F7040/8040 type broiler cage is a stacked cage design inspired by broiler brooding cage with air passage. It consists of a cage system, a feeding system, a drinking water system, and a removal system. The automatic control of feeding, drinking water and defecation of large-scale broiler farming has been realized, which greatly saves labor cost and time.

PARAMETERS

- Unit length: 1400mm/1250mm
- Cage width: 2000mm
- Air passage width: 400mm
- Cell height: 450mm
- Single floor height: 590mm
- Three floors height: 2030±20mm
- Four floors height: 2620±20mm
- Unit cage dimension(LxWxH):
1400mm/1250mm × 700mm/800mm × 450mm
- Unit cage drinker quantity: brooding floor 3 pcs, the other floors 2 pcs
- Door Dimension: 1370mm/1200mm × 270mm (horizontal type)
- Unit cage breeding capacity: 20 birds (2.5kg/bird)
- 3 tier breeding capacity: 20 × 6 = 120 birds (2.5kg/bird)
- 4 tier breeding capacity: 20 × 8 = 160 birds (2.5kg/bird)



PRODUCT ADVANTAGES

- 400mm width air passage in the middle remarkably improves ventilation
- Open longitudinal support design reduces the obstacles of lateral ventilation in the cage and improves the ventilation advantage in the cage;
- Longitudinal support is integrated with the chick pedal with high strength;
- Open inter manure removal system, easy to install, observe, and rinse the cage;
- Anti-deviation device is equipped at the head and end, effectively reduce the deviation of the manure belt;
- The rear end is equipped with double scrapers, and the scraping effect is good. The tail scraper is convenient for disassembly and maintenance.
- Two brackets are arranged at the bottom of each bottom net, and the strength of the bottom net is better;
- In-line push-in large cage door, convenient for catching chickens, and low rate of chicken disability;
- 360° triggered cone drinker, good sealing and easy to flush;
- Double side troughs and a large deep slope trough to reduce the feed spread of the birds.



Cage body



Cage front



Cage rear

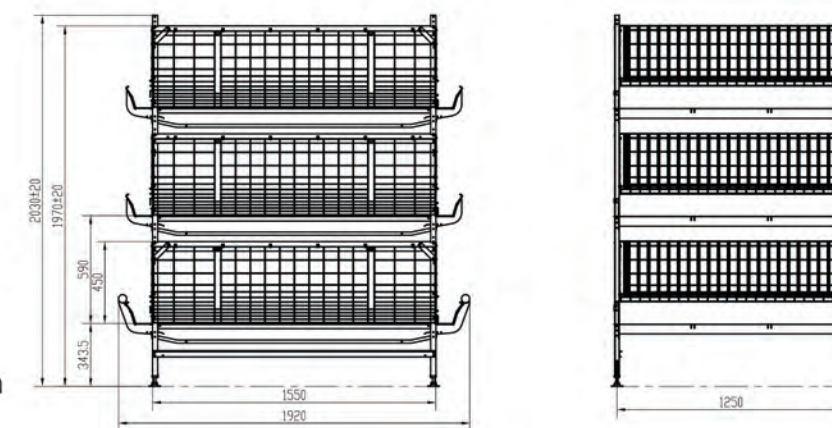


EI-LY/J ECONOMY BROILER CAGE

EI-LY/J15 is multi-tier cage system for broiler feeding, composed of cages, drinking system, feeding system, manure removal and discharge system. It realizes large-scale control of drinking, feeding and manure removal, saves labor cost and time.

PARAMETERS

- Unit length: 1250mm
- Cage width: 1550mm
- Cell height: 450mm
- Single floor height: 590mm
- Three floors height: 2030±20mm
- Four floors height: 2620±20mm
- Unit cage dimension(LxWxH):
1250mm×775mm/800mm×450mm
- Unit cage drinker quantity:
brooding floor 3 pcs, the other
floors 2 pcs
- Door Dimension:1200mm×270mm (horizontal
type)
- Unit cage breeding capacity:19 birds (2.5kg/bird)
- 3 tier breeding capacity:19×6=114 birds
(2.5kg/bird)
- 4 tier breeding capacity:19×8=152 birds
(2.5kg/bird)



Small crane type traveling hopper feeder



Push type door



Manure belt longitudinal bracing

PRODUCT ADVANTAGES

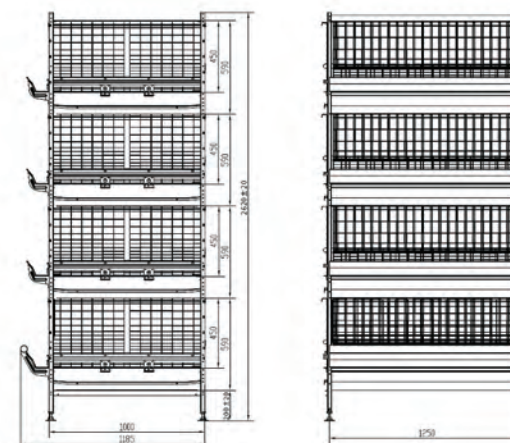
- Open longitudinal support design reduces the obstacles of lateral ventilation in the cage and improves the ventilation advantage in the cage;
- Longitudinal support is integrated with the chick pedal with high strength
- Open inter manure removal system, easy to install, observe, and rinse the cage
- Anti-deviation device is equipped at the head and end, effectively reduce the deviation of the manure belt;
- Two brackets are arranged at the bottom of each bottom net, and the strength of the bottom net is better;
- In-line push-in large cage door, convenient for catching chickens, and low rate of chicken disability;
- 360°triggered cone drinker, good sealing and easy to flush;
- Double side troughs and a large deep slope trough to reduce the feed spread of the birds.





EI-LY/DC1000 ONE SIDE FEEDING BROILER CAGE

EI-LY/DC1000 One side feeding broiler cage is a multi-tier cage system for broiler feeding, composed of cages, drinking system, feeding system, manure removal and discharge system. It realizes large-scale control of drinking, feeding and manure removal, saves labor cost and time.

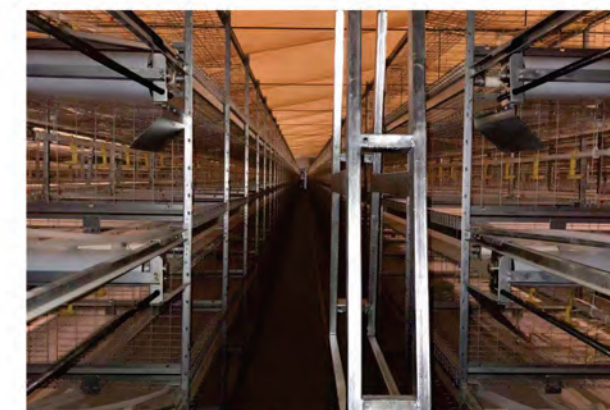


PARAMETERS

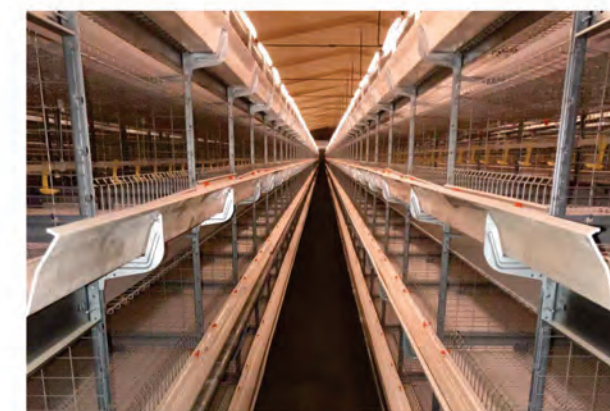
- Unit length: 1250mm
- Cage width: 1000mm
- Cell height: 450mm
- Single floor height: 590mm
- Three floors height: 2030±20mm
- Four floors height: 2620±20mm
- Unit cage dimension (LxWxH): 1250mm×1000mm×450mm
- Unit cage drinker quantity: 3 pcs
- Feeding side door dimension: 1200mm×270mm (push type)
- Harvest side door dimension: 1200×450mm (up and down type)
- Unit cage breeding capacity: 25 birds (2.5kg/bird)
- 3 tier breeding capacity: 25×3=75 birds (2.5kg/bird)
- 4 tier breeding capacity: 25×4=100 birds (2.5kg/bird)

PRODUCT ADVANTAGE

- 1000mm cage width, better vertical ventilation;
- Open longitudinal support design reduces the obstacles of lateral ventilation in the cage and improves the ventilation advantage in the cage;
- Longitudinal support is integrated with the chick pedal design can take human weight when catching chicks;
- Open inter manure removal system, easy to install, observe, and rinse the cage;
- Anti-deviation device is equipped at the head and end, effectively reduce the deviation of the manure belt;
- Two brackets are arranged at the bottom of each bottom net, and the strength of the bottom net is better;
- Push up and down the split type large cage door, the operation space is large, the chicks are convenient to be caught, and the chicken disability rate is low;
- 360°triggered cone drinker, good sealing and easy to flush;
- Double side troughs and a large deep slope trough to reduce the feed spread of the birds.



Seeder type traveling hopper feeder



Deep slope trough



Push type door



Anti off-track mechanism

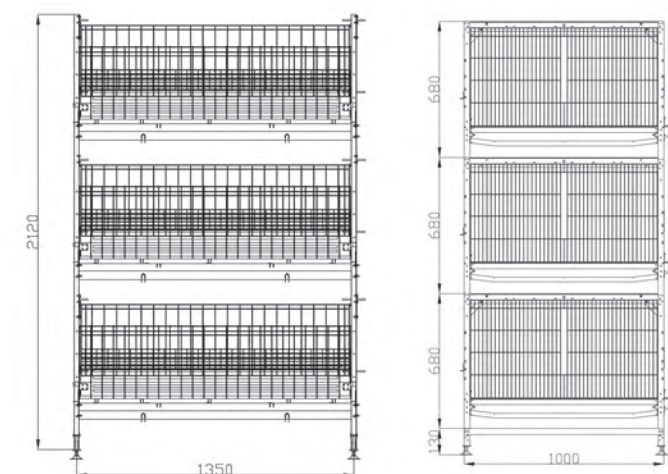




DUCK CAGE

EI-LY/RV13510 DUCK CAGE

1 m wide duck cage is multi-tier cage system for duck breeding, composed of cages, drinking system, lighting system, manure removal system, etc. The cage width can be customized between 1m~1.5m.



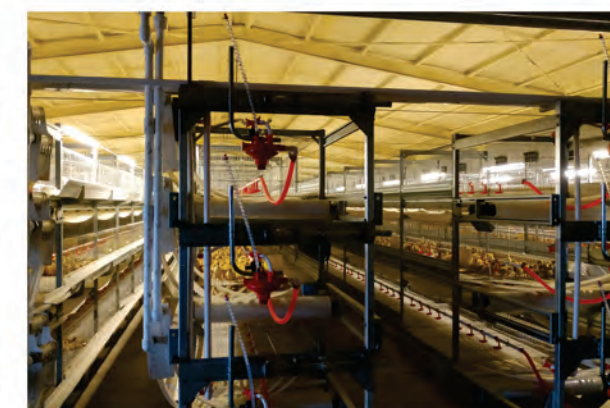
EI-LY/RV13510

PARAMETERS

- Tier height: 680mm
- Cell height: 530mm
- Unit length: 1350mm
- Cage width: 1000mm
- 3-tier cage height: 2190±20mm (4-tier as optional)
- Door dimensions: swing up door dimensions in duck catch side 470mm×1325mm, include manure board. Mesh dimension in trough side 370mm
- Each unit tier can breed 22 birds (3kg/bird)
- 4 drinkers in each unit tier
- Adjustable height of drinking line: 100mm-435mm
- Composed with seeder type traveling hopper feeder, travel speed 3m/s, each hopper holds 250kg feed.

PRODUCT ADVANTAGE

- 360°triggered cone drinker, good for duck drinking and easy to clean;
- Swing up door, easy to catch birds;
- Cage body is reinforced with longitudinal brace, high strength and effectively solve the problem of belt off-track;
- Seeder type traveling hopper feeder with feed uniform device;
- Manual bird harvest.



Cage front



Cage rear



Drinking



Brooding



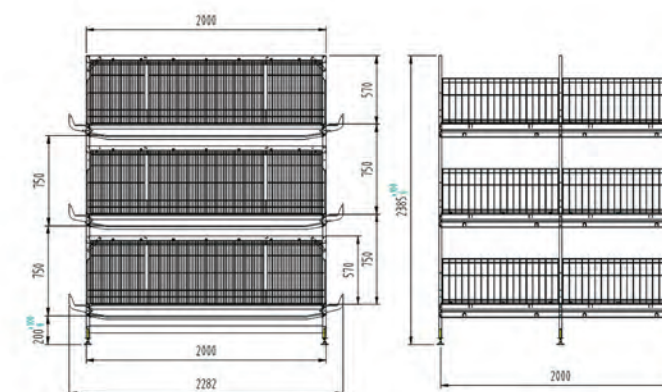
EI-LY/RV20020 DUCK CAGE

2 m wide duck cage is multi-tier cage system for duck breeding, composed of cages, drinking system, lighting system, manure removal system, etc. The cage width can be customized between 1m~1.5m.



PARAMETERS

- Tier height: 800mm
- Cell height: 570mm
- Unit length: 2000mm
- Column interval: 1000mm
- Cage width: 2000mm width including trough 2300mm
- 3-tier cage height: 2600±20mm 4-tier as optional
- Bird No in each unit tier ≤ 64 (3kg/bird)
- Trough configuration: 2 lines for each tier (on both sides)
- Drinking line configuration: 2 lines in each unit tier.
- Drinker qty: 10 pcs in each unit tier.
- Drinking line adjustable height: 110mm~470mm
- Travel speed of manure belt: 3m/min



EI-LY/RV20020-3

PRODUCT ADVANTAGE

- The frame structure consists of 'longitudinal support + vertical column + diagonal support', with high longitudinal and cross strength. The bottom mesh is strengthened by cross bracing with special section.
- The trough side is composed with duck baffle plate and baffle net. Heightened duck baffle plate to prevent duck from escaping and feed leakage. Baffle plate is easily adjustable making it harmless to ducks.
- The heightened trough can prevent spillage and avoid feed waste.
- Swing up door is convenient to open and close, catch duck, and do injection.
- More intensive separate mesh can effectively avoid duck shuttling between cells; the opening at the bottom of the mesh is low.
- The drinking line can be lowered to a very low level where the ducklings can easily drink; Stainless steel hang rope has excellent wearability. The white color rope buckle can prevent rope from being loose caused by duck pecking.
- The special anti-off-track device in the rear and the angled manure belt bracket would greatly reduce the deviation of manure belt.
- The auger type manure discharge system is good at cleaning the thin duck manure.
- Manual duck catch on both sides.



Duck catch on both sides



Cage front



Illuminate each other



Dedicated drinker for duck



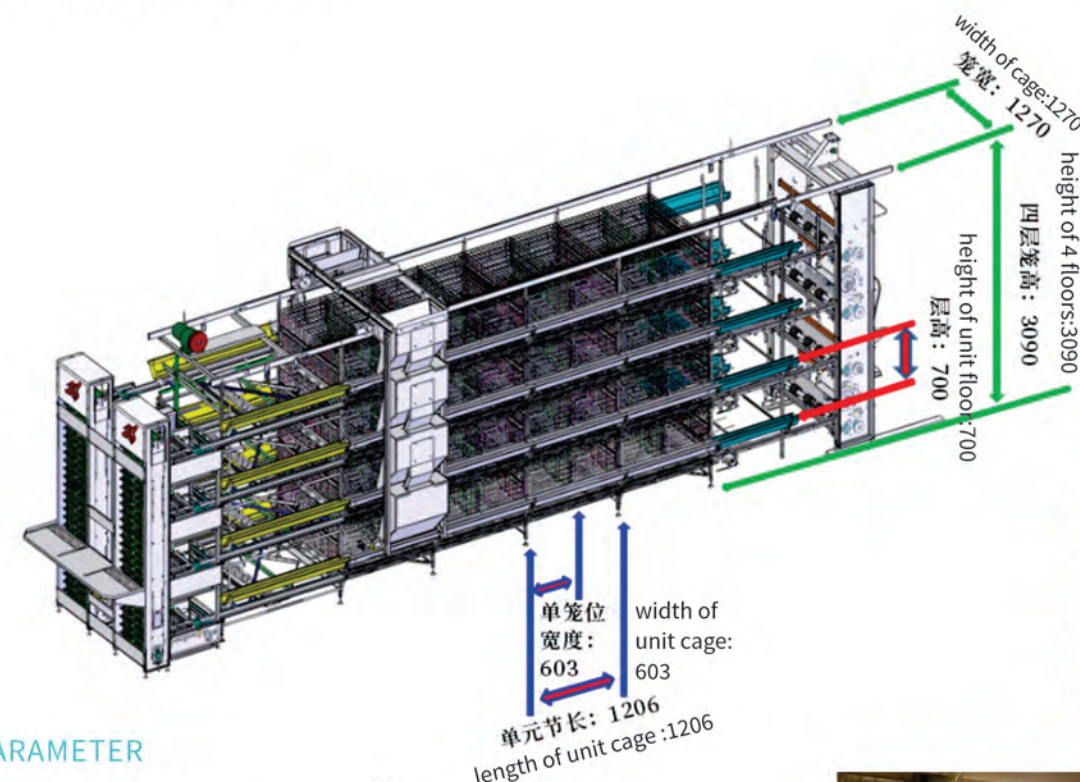
LAYER CAGE

Now China poultry breeding industry is in transition to large-scale, intensive, specialized and standardized. Without R&D of automatic and intelligent breeding equipment, industrial upgrading cannot be achieved because of the elevated market concentration. H type cage undoubtedly meet the needs of the industry.

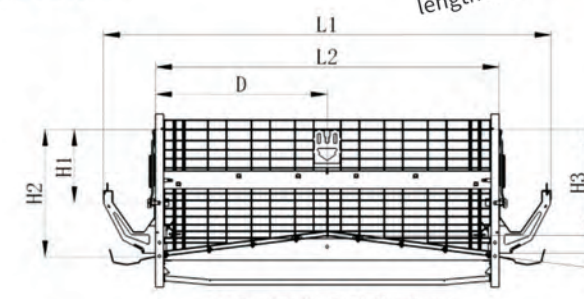


TRAVELING HOPPER CAGE

Traveling hopper cage features with low energy consumption and low failure rate; feed tank and feed trough not touching effectively avoids the wear of feed trough; feed regulator with V-shaped trough can precisely control feed, improve the uniformity of the layers.



PARAMETER



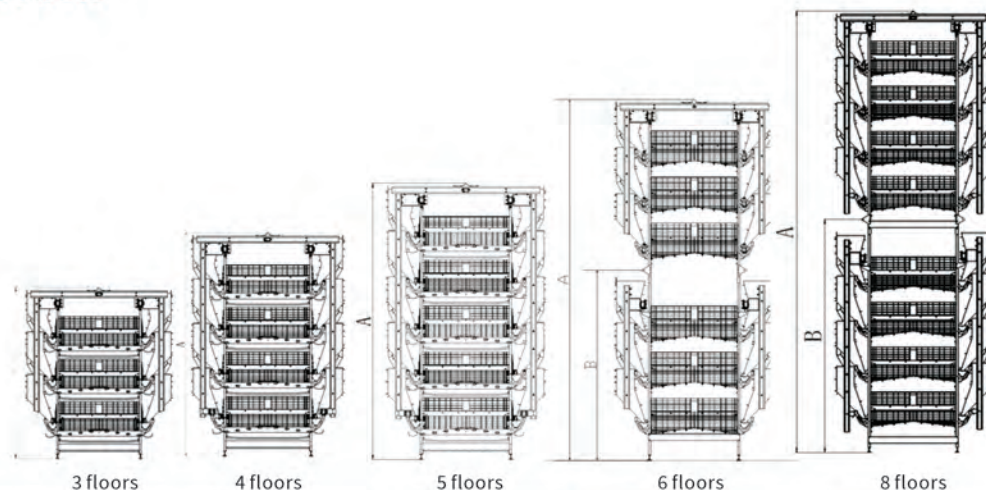
行车式H型商品蛋鸡笼
H type layer cage—traveling hopper



Cage Parameter	Unit	H type layer cage—traveling hopper
(Standard height/tier)	mm	700
{Unit depth D}	mm	635
(Unit width L)	mm	603
(Unit door height H1)	mm	236
(Unit front height H2)	mm	473
(Unit rear height H3)	mm	397
(Bottom mesh sqcm S)	cm ²	3829
(Bottom mesh angle α)	°	7
(Number of birds/unit)	bird/unit	9
(Space/bird S)	cm ²	425
(Cage width L2)	mm	1270
(System width L1)	mm	1655



Main Models



Floor	3	4	5	6(3+3+walkway floor)	8(4+4+walkway floor)
A	2630mm	3270mm	3910mm	5031mm	6311mm
B				2580mm	3220mm

Systems

Egg Collecting System

Automatic egg collecting system can improve working efficiency and reduce the risk of disease transmission

- Collecting capability: 12,000 pc/h
- Work smoothly, breaking rate lower than 5‰
- Belt-type egg collecting machine has a simple structure, and is designed with a belt tensioning and correcting device, which is convenient to use and maintain.



Cage Front

- Belt adjustment method adopts the chain + bolt structure, which can realize quick and precise adjustment



Cage Body

- Bottom Net Dimension : 25.4mm×38.1mm
- Net Diameter: 2.0mm
- Bottom mesh slope is 7° to ensure smooth egg rolling. Unique bottom mesh installation method (soft and tightness adjustable) has a high quality trampoline effect, that reduces the egg breaking when falling and rolling.



Top Mesh

- Each floor is equipped with a top mesh to prevent the layers from licking the upper dung belt and reduce the risk of cross infection





Egg Collecting

- Efficiently filter soft shell eggs and contaminants to prevent soft shell eggs from contaminating other eggs in the ladder collecting
- Adjust eggs and deliver them to the ladder collecting system in the best posture
- Adjust eggs running direction of each floor so that the eggs in each column are evenly distributed to the ladder collecting system to avoid collision and reduce the egg breaking



Egg Buffer

- Egg Buffer effectively reduces the probability of breaking eggs caused by collision when eggs roll from cage to egg belt.
- Moisture from new eggshell evaporates immediately, reduce the dust and layers feathers
- Egg picking time and period can be set based on user needs.



PVC Shielder

- Effectively prevent layers dung from falling into the under floor of egg belts and troughs, avoiding contamination of eggs and feed
- Effectively prevent the layers from licking and stepping on the eggs, reduce egg breaks

Sliding Cage Door

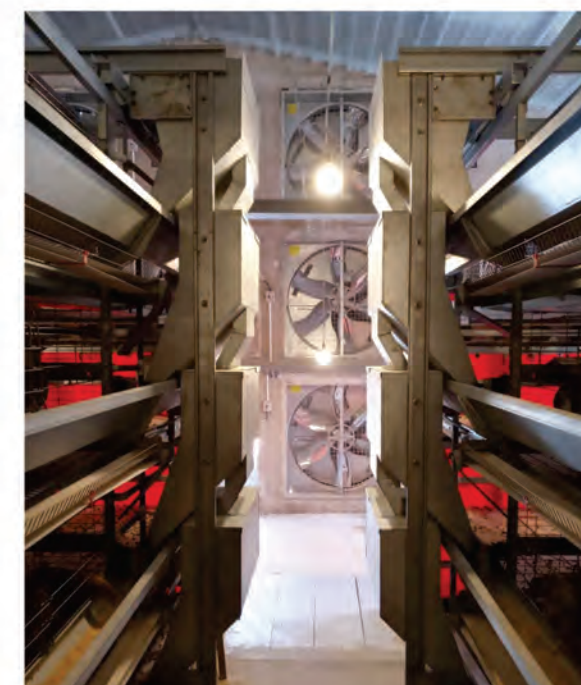
- Sliding cage door uses the trough as a glide path, which is light in operation and does not hurt the layers in daily work.

Traveling Hopper

- Traveling hopper is fully sealed off.
- Hopper structure prevents feed clogging and component separation.
- Hopper capacity is 90L (about 60Kg), and feeding four times in a day.
- Feed regulator ensures high precision feeding, and solves the problems of nutrient loss caused by long stock time and poor evenness of layers caused by uneven feeding.

Feed Trough

- Seamless butt joint design of the feed trough not only ensures that there is no deteriorated material at the bottom, but also can be used as the track of the driving and observing vehicle. Feed trough is free from wear and tear because the feed trough does not touch the feed trough. Feed trough no damage because of no touch with homogenizer.
- Feed trough with large gradient structure can avoid the waste of feed caused by layer eating. The trough structure has high strength, and the staff can trample on it to observe the layers.





Drinking Line

- Drinking lines are installed in the middle of the cage
- V-shaped water tank can prevent water from falling down to belt.
- Automatic reverse flushing and water volume detection devices improve production efficiency.



Automatic Feed Recycle System

- Automatic feed recycle system is at the front and end of cage body. It effectively collects feed in the no-layer area, prevents feed degeneration and saves labor.

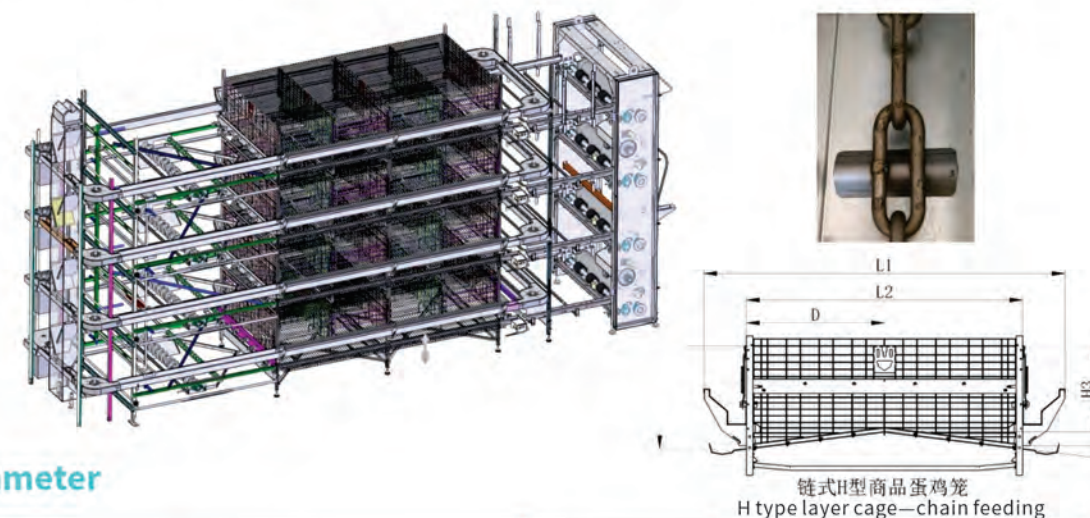


Cage Rear

- We use secondary PU material scraping board to clean the dung belt. This avoids mutual infection caused by dung with less damage to the belt itself, ensuring the service life of the belt
- Multi-layer scraper adopts linkage mechanism

Chain Feeding Cage

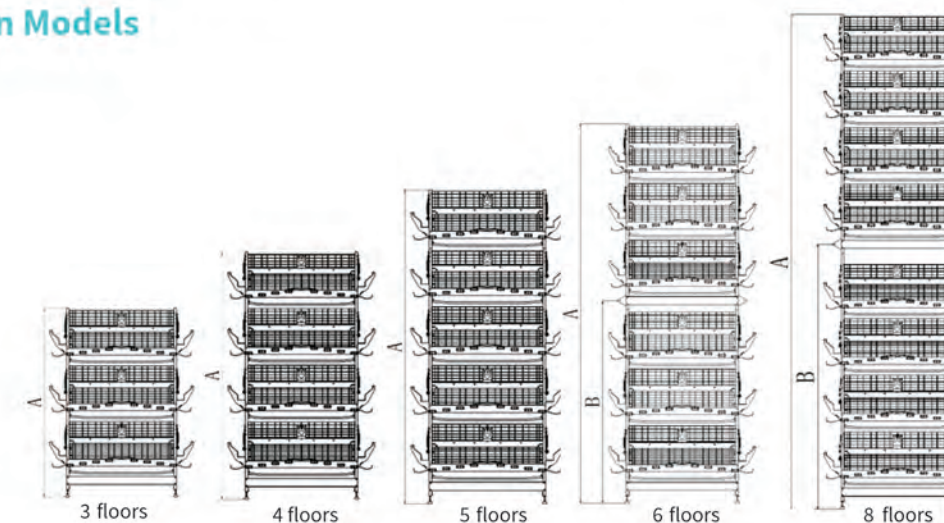
Chain feeding features with fast feeding speed, which can renovate feed at the bottom of feeding trough and improve the feeding quality of layers. Each feeding line can be operated independently without affecting the feeding of layers in the process of operation, which is flexible and efficient. Our self-developed chain is stable in operation with low failure rate.



Parameter

Cage Parameter	Unit	H type layer cage—chain feeding
(Standard height/tier)	mm	700
(Unit depth D)	mm	635
(Unit width L)	mm	603
(Unit door height H1)	mm	236
(Unit front height H2)	mm	473
(Unit rear height H3)	mm	397
(Bottom mesh sqcm S)	cm ²	3829
(Bottom mesh angle α)		7
(Number of birds/unit)	bird/unit	9
(Space/bird S)	cm ²	425
(Cage width L2)	mm	1270
(System width L1)	mm	1660

Main Models



Floor	3	4	5	6(3+3walkway floor)	8(4+4+walkway floor)
A	2146mm	2786mm	3426mm	4268mm	5586mm
B				2280mm	2990mm



FLOOR FEEDING EQUIPMENT

- Floor feeding equipment is mainly used for automatic control of feeding, drinking, feed supplying, ventilation, heating and humidification in poultry house. It carries out unified and standardized feeding management for common poultry and small livestock, and construct large-scale intensive and green ecological scale feeding mode.
- Floor feeding equipment cannot only maximize the internal resources of poultry houses. It also reduce feed waste, save cost, and also improves the density of feeding, reduce labor intensity and force, reduce the probability of disease transmission and adapt to the modern poultry industry information management requirements.

◆ FEEDING SYSTEM

◆ DRINKING SYSTEM

◆ PLASTIC SLAT





FLOOR BREEDING EQUIPMENT

FEEDING SYSTEM

The feeding system is a fully automatic feeding solution for chicks, cocks, ducks, geese, and rabbits.

FEEDING LINE



TECHNICAL INDEX

- It is suitable for day-old chicks, no need for additional feeders for young chicks.
- Feed intake needs from day old chicks to adult birds can be guaranteed by simply adjusting the height of the feed line.
- Reduces feed waste.
- The rotating drive winch is labor saving since it can adjust the whole feeding line to the appropriate feed height.



MAIN FEATURES OF EI-WL2/14 FEED PAN

- 14 feeding positions, suitable for chicks from day old to adult.
- There are four blanking windows at the lower part of the feed pan rack.
- The adjusting sleeve can accurately adjust the height of feed pan.
- The shed can be effectively sectioned with movable imported feed position detection switches.
- The shell and chassis are screwed up for safe and convenient installation and disassembly.
- Bright cover, easy to clean and no dead angle.



MEAT DUCK FEEDING LINE



TECHNICAL INDEX

- It is suitable for day-old ducks, no need additional feeders for baby ducks
- Feed intake needs from one day old ducks to adults can be guaranteed only by adjusting feed line to an appropriate height.
- Reduces feed waste
- Labor saving. Rotating drive winch can adjust the whole feeding line to the appropriate feed height.



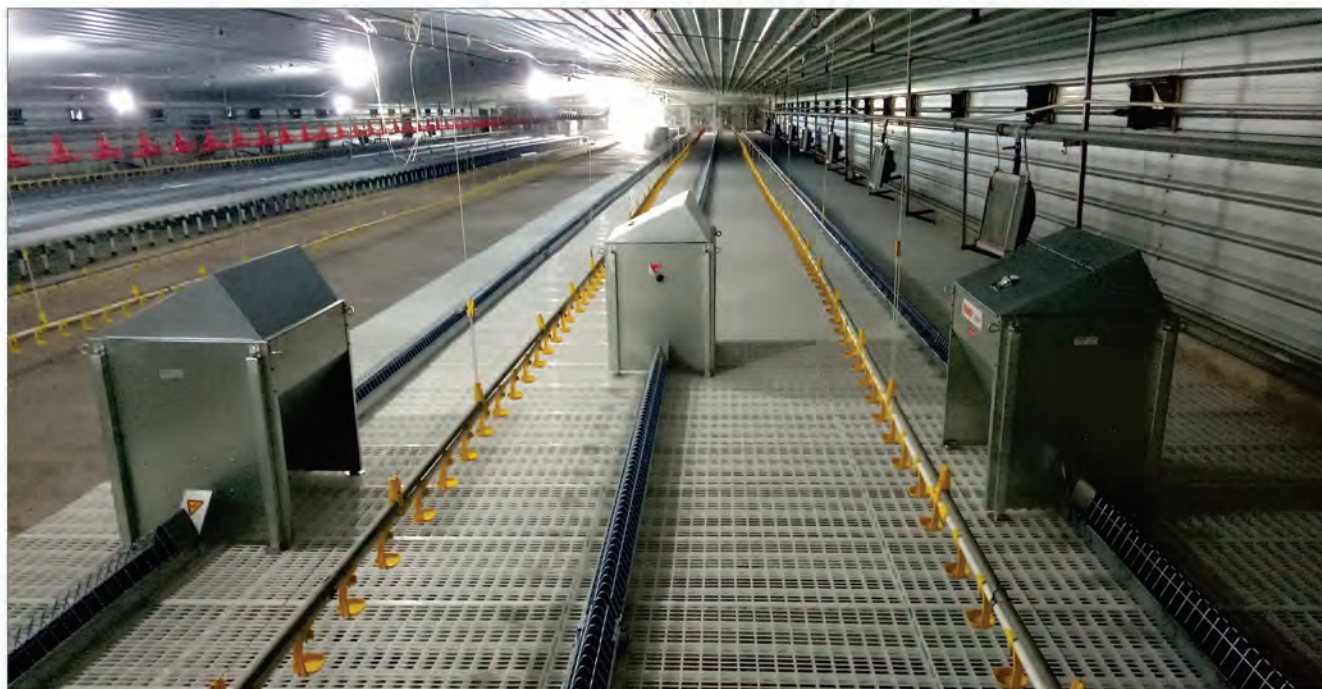
MAIN FEATURES OF EI-WL3/5 FEED PAN

- The feed pan can be used for duck, rooster and rabbits
- The shell with five grills is designed with large space to meet the requirement of big size poultry.
- The adjusting sleeve can accurately adjust the height of feed pan
- There are four blanking windows at the lower part of the feed pan rack which is suitable for day-old ducks feeding.
- The chassis is designed with double trough, which can facilitate the free access of ducks to avoid feed waste.
- The shell and chassis are hooked together allowing a safe and convenient installation and disassembly.



BREEDER FEEDING LINE

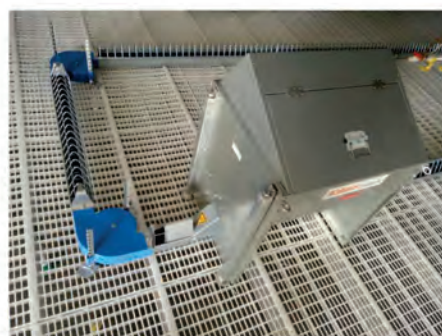
CHAIN FEEDING



Chain feeding system has superior performance recognized in the poultry industry. It has the advantages of uniform feeding at a fast pace. The length and width can be customized in accordance with the envisioned poultry house to meet the diversified needs of customers.

TECHNICAL INDEX

- It meets the function of cycle feeding. The size of the fabric in the feed port can be adjusted.
- The feeding trough is reasonable in structure and can meet the needs of adult egg breeders and broilers to peck the feed without injuring the comb and beak; the trough is connected by plug-in joint.
- The surface of the trough is smooth, rust resistant, corrosion resistant and easy to clean.
- The chains are rust resistant, wear resistant and anti-corrosion. They have enough strength and stiffness to endure constant movement of its parts during operation. The chain structure is unidirectional, driving the feeds only in one direction.
- The electric drive wheel and the chain piece are properly meshed, providing a flexible drive movement and avoids the chain from slipping. The driving motor has a safety insurance device that avoids the feed line from jamming.
- Both sides are equipped with a 90 degree angle structure to ensure a stable and flexible steering and transmission of the chain piece.



- This system is composed with support components, protective components, limit grids, feed hoppers and other components, which are easy to install, clean and maintain.
- The feed hopper is larger than 300l, ensuring that there is enough feed at all time, and it's equipped with support feed hopper.
- The limit width of chain feeding system is 45mm, which is suitable for chicks eating and prevents chicks from scrambling.
- The chain running speed is 18-36m/min, conveying speed is fast, feeding uniformity is excellent, the maximum feeding capacity is not less than 1.8t/hour.

FEED PAN AND AUGER FEEDING



Feed pan and auger feeding system is one of the feeding methods for breeder. Its circle feeding mode is suitable for the automatic feeding of hens when male and female are raised together.

MAIN ADVANTAGES



- It is suitable for all kinds of hens feeding. It can adjust the width of the feeding window according to the size of the hen head without using any tools.
- It can reduce feed waste. The chassis is designed with patented technology. When chicks are eating, feed is concentrated in the chassis flash plate, far from the edge, feed is rarely thrown out of the plate by the chicks, which effectively prevents feed waste.
- It can save labor. The whole feeding line is suspended on the beam, and the staff can adjust the feed pan of the whole feeding line to the appropriate height by rotating and driving the winch.

MAIN FEATURES OF EI-WL1/16.2 FEED PAN

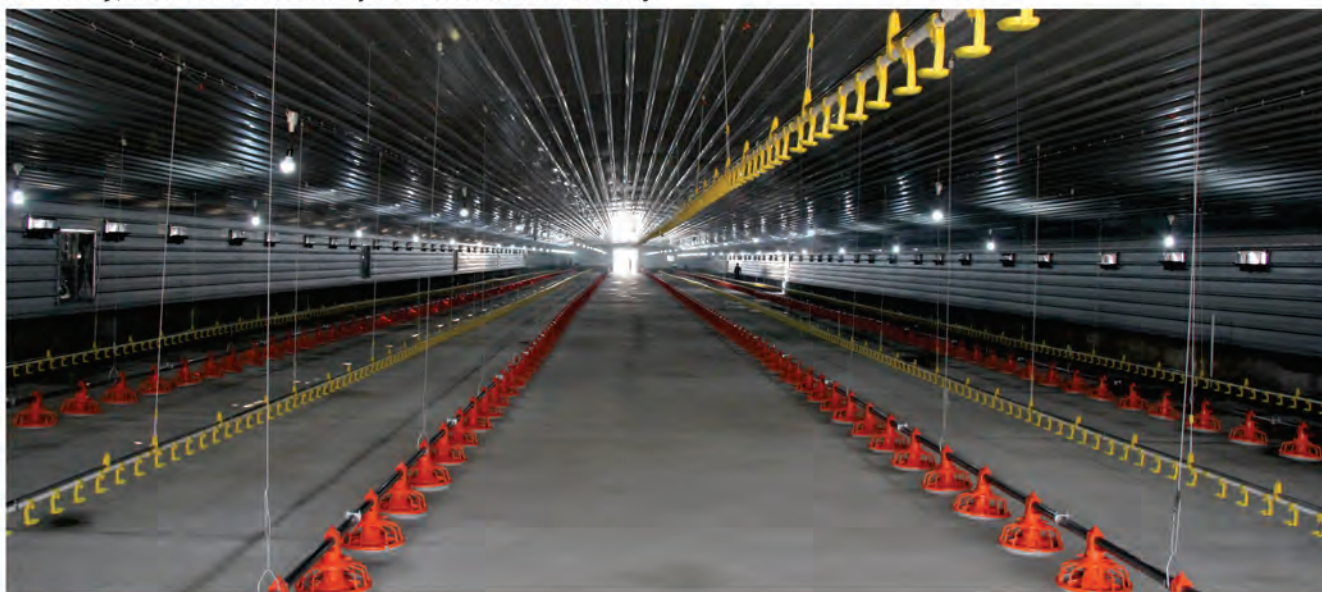
- 16 feed positions.
- The feed pan is designed for anti-corrosion, anti-aging, anti-ultraviolet, and can resist the erosion of various common cleaning agents and disinfectants.
- Unique adjustable design that can adjust the size of the feeding window according to number instruction, while using up and down positioning, and the size of the feeding window will not change because of the cocks crowding and grabbing. After the window is adjusted, only hens can eat feed.
- The chassis is designed with double trough, which can facilitate the free access of young chicks to avoid feed waste.
- By adjusting the stock in the feed pan, the feed can be quickly filled to ensure that chicks which at different positions in the poultry house can eat the feed at the same time.
- The shell and chassis are hooked together allowing a safe and convenient installation and disassembly.





DRINKING SYSTEM

The Ei Farming Nipple drinking system provides the right amount of clean water for the proper age of the birds without leakage. This is complemented with sufficient filtration, pressure regulation, medication/dosing device, and volume flow measurement. The basic Ei Farming auto drinking system is composed of the nipple drinkers on a drinking line, pre-treatment assembly, water regulators, suspension assembly, anti-roost assembly and terminal assembly.



MAIN FEATURES

- Nipple drinkers are properly sealed to maintain a dry floor for the birds.
- It supplies bacteria-free drinking water for the birds.
- As the birds grow, the height of the drinking line is easily adjusted without the need for complex tools.
- The water flow for each nipple drinker is precisely controlled by the pressure-reducing valve for different stages of poultry growth.
- There is a variety of easy to install water-drip plates to choose from.
- The drinking line tubes and supporting steel pipes are of high strength and are securely anchored against possible collisions from adult birds and/or flock men.

PRETREATMENT SYSTEM FOR DRINKING LINE



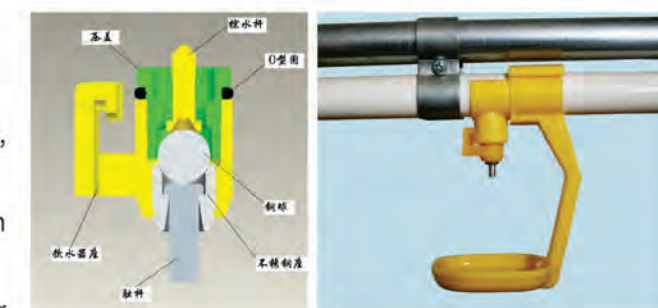
- The system includes a series of filters, water meters, medicator/dosing device and maintenance valves. The combination of these vital elements ensures that the water in the line is fresh and clean for the birds to drink.
- Filter: The reverse-flush filter purifies the water making it fit for drinking and prevents the nipples and water lines from being clogged with sediments.
- Electronic water meter: This can be interfaced with the controller to show an accurate volume of water consumption for periodic measurement based on the growth rate and/or age of the birds.
- Medicator: Any change in the flow or pressure from the water source does not affect the desired proportion of medicine that needs to be combined with the drinking water.

NIPPLE DRINKER

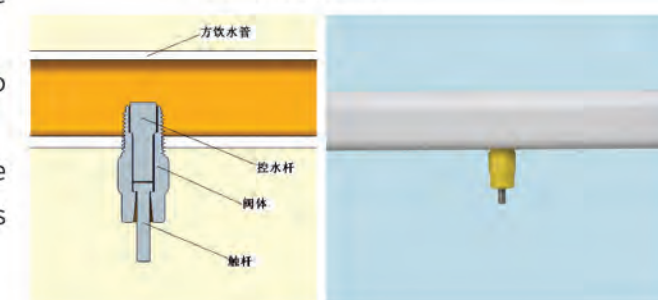
BALL VALVE DRINKER

MAIN FEATURES

- High quality stainless steel, three layer seal, reliable performance.
- The 360 degree nipple ensures poultry drinking in all direction.
- The young chicks can drink by pecking side or jacking the valve, these two ways can both provide adequate water.
- High sensitivity, small light trigger force and no leakage.
- The high precision water control pole can provide different kinds of water dispenser which has different water outflow for various poultry.



Ball valve drinker



Cone valve drinker

CONE VALVE DRINKER

MAIN FEATURES

- High tightness: The water control rod and valve core are machined by super high precision numerical control vehicle to ensure that the cone valve drinker has good sealing performance.
- Small light trigger force: In the range of 150mm-200mm water pressure column, the triggering force is small, which is benefit for baby chicks drinking.
- Good resistance: It is imported copolymer formaldehyde resin injection valve body with high strength, good wear resistance and it's suitable for chicks and ducks drinking needs.
- Small lateral trigger flow: In young chick and duck breeding stage, the side light trigger force of cone valve water dispenser is small within the range of 50mm-200mm water pressure column, and the flow is also small, which is suitable for chicks and ducks drinking.
- Large top flow: At normal temperature, the water pressure 50mm-220mm, water level 400mm-420mm, the flow rate is 30-150ml/min, suitable for broilers, meat-type ducks, breeders of chicks and ducks in different seasons.
- Good clogging resistance: It is processed by super high precision machining. The surface of components is clean and not easy to jam.



Chicken Drinking Water Illustration



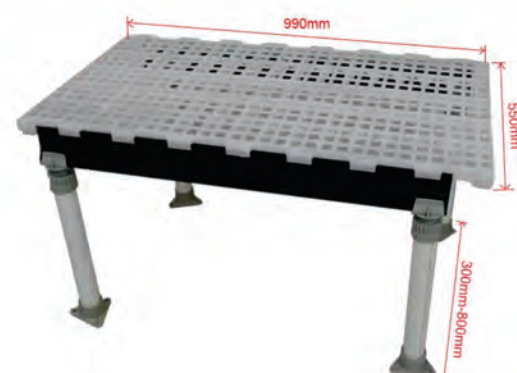
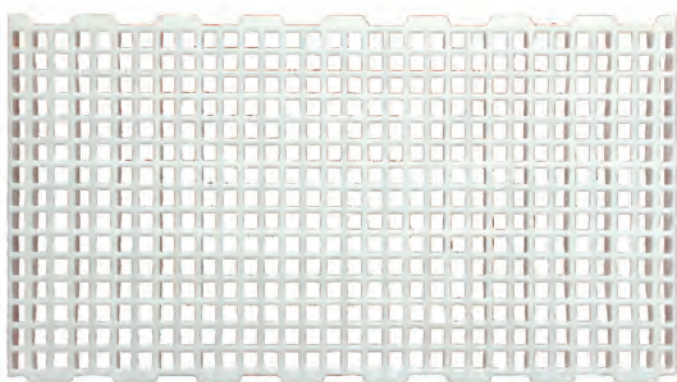
Duck Drinking Illustration



PLASTIC SLAT

Plastic floors are generally laid in breeder houses, net chicken house or duck houses. Chicks and ducks live on the floor surface. There are meshes on the plastic slat. The manure falls under the plastic slat to maintain a good growth environment for poultry.

EI-DB/S99 TYPE PLASTIC SLAT

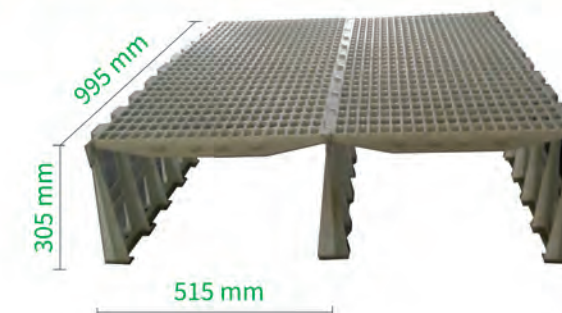
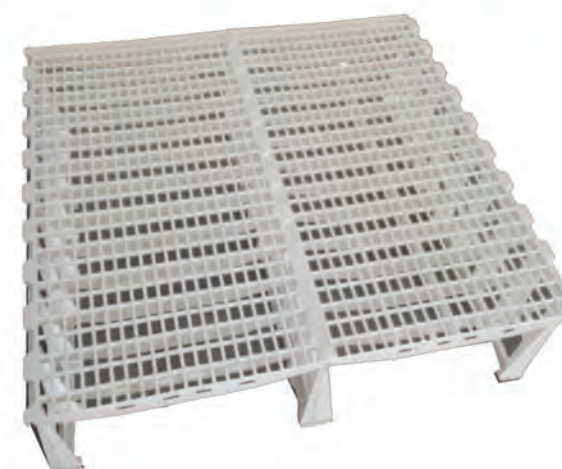


PRODUCT ADVANTAGES

- Average bearing capacity: 100kg/m²
- Leak hole size: 24mm×19mm
- Plate size: 990mm×550mm×40mm
- Adjustable height range: 300~800mm
- Usage: Breeder, broiler, duck house
- Features: Easy cleaning, fast installation, long lifespan.
- Design: Arc design is harmless to poultry legs, good-looking and practicable



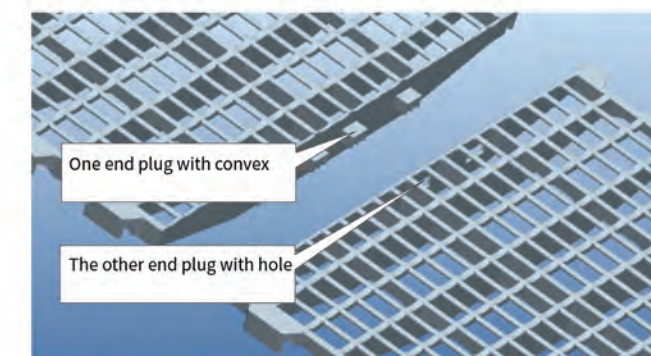
EI-DB/S99X TYPE PLASTIC SLAT



The dimension of each slat: L*W*H=995x515x48mm
 The shape and size of the plastic floor are as follows
 Each piece has a flat area of 0.5m²
 The size of fixed leg is 995mm long and 305mm high.

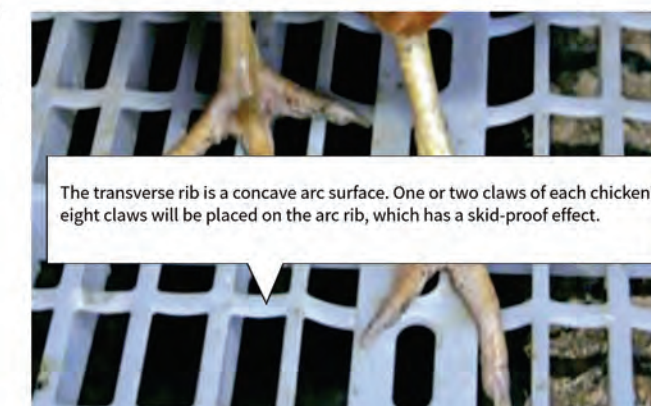
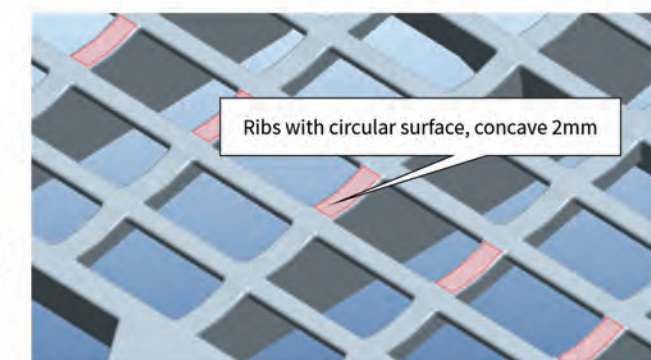
MAIN FEATURES

- Plastic slat is made of polypropylene:
- Good corrosion resistance: resistant to manure and general disinfectant.
- High strength: 5mm concave at 60kg loads.
- Good low temperature toughness and fall resistance.



PRODUCT ADVANTAGES

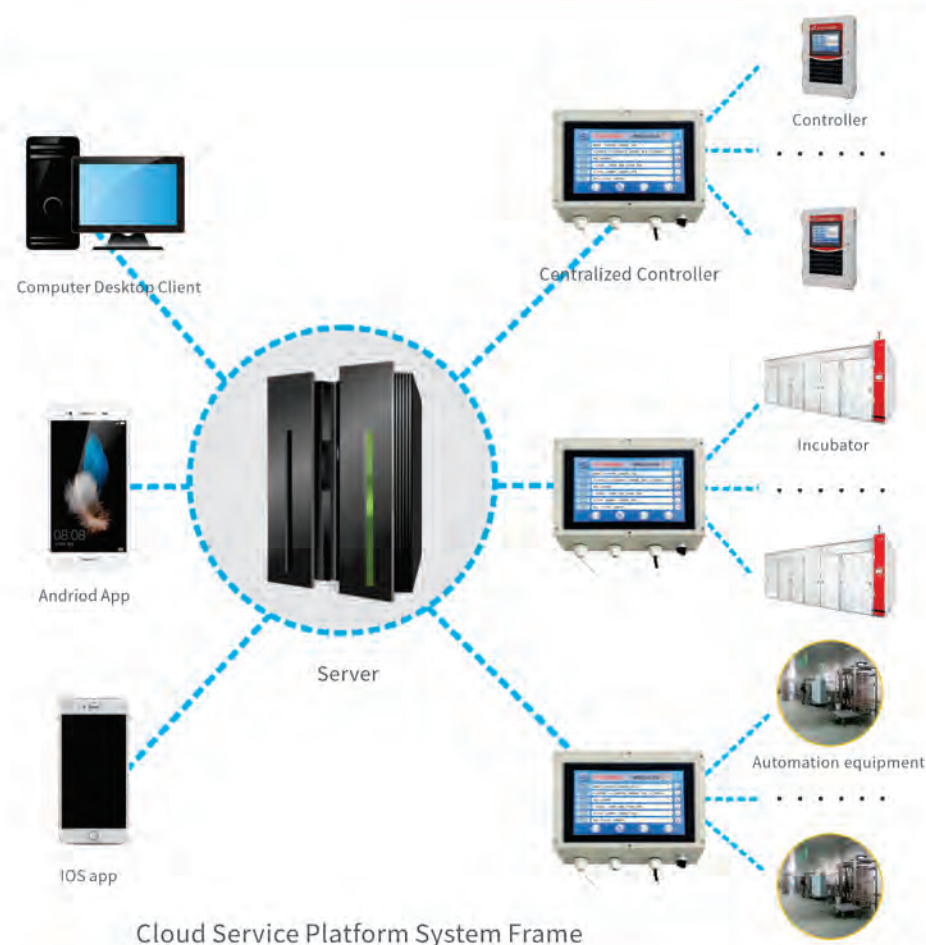
- Increase the insertion of both ends to ensure the flatness and integrity of the slab.
- Increase the size of leaking hole, the area of leaking can reach 57%, which is convenient for leaking and cleaning.
- The transverse ribs are designed by concave arc, and the anti-skid design ensures that the breeders do not slip during mating.
- The surface edges of the slat are all circular arc structures to protect breeder's feet from injury.



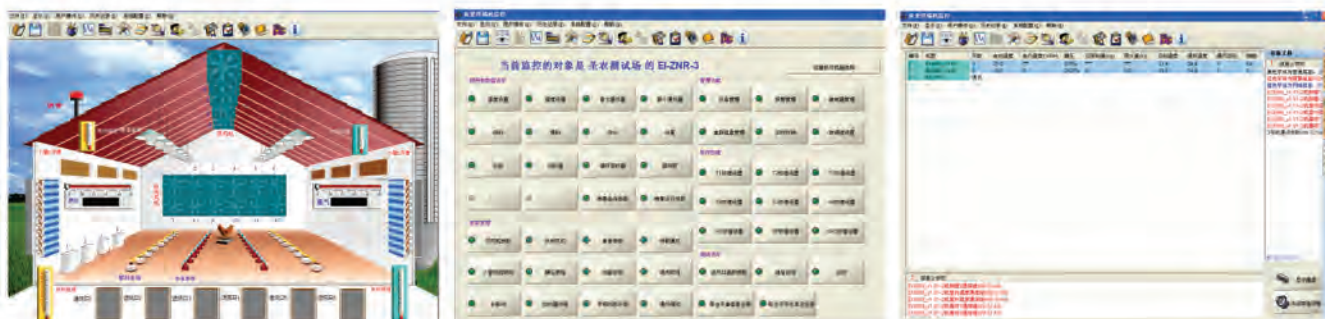


EI BREEDING CLOUD SERVICE PLATFORM

EI breeding cloud service platform connects EI series environmental control equipment in the farm to the same network, realizes dynamic management and centralized monitoring of the equipment, and uses cloud server, centralized controller (local server), Windows client, Android APP monitoring to realize local and mobile management of the breeding process.



Cloud Service Platform System Frame



Centralized controller is the core equipment, which connects many production equipment and upload data information to the cloud. When separated from the cloud it can be used as independent operation of local systems, and it can provide local monitoring services; When connected to the cloud it can provide basic data information for the cloud. Data analysis is done on the cloud to form an expert system, which guides users to set production process parameters in turn.

It can control and manage the real-time status of various production equipment, environmental control equipment, such as switch, bug, abnormal power failure and communication failure, alarm information. It also notice of warning and maintenance prompt for equipment and material link ,and it has the functions of unified maintenance of process parameters and view of historical curve. Breeding cloud service platform provides different access rights for local staff on duty, field managers and business managers to meet the customized needs of different enterprises.

PRODUCT ADVANTAGE

- **Server + Client.** The system separates the storage of data from the user's operation. The server boots 24 hours to record the running data of the device and provides status notification service. Client opens at any time according to user's needs, supports multi-user login at the same time, supports windows, android, iOS system client, to meet more user needs.
- **Real-time fault alarm board.** It can display alarm and abnormal information of all fault devices in real time (abnormal power failure, communication failure).
- **Curve analysis.** The system records the running data of the equipment regularly and presents it in various forms such as tables, curves and so on. When displaying the curve, the fault points are displayed on the curve, and the data near the fault points are amplified and analyzed.
- **Guidelines for troubleshooting.** When the equipment fails, the possible cause of the failure will be prompted to the user, and guide the user to measure the signal on the equipment for troubleshooting.



Mobile Client



Computer Desktop Client



POULTRY FARM PROJECT

- ◆ STEEL STRUCTURE POULTRY HOUSE
- ◆ FEED SUPPLYING SYSTEM
- ◆ HEATING SYSTEM
- ◆ PROJECT CASE

Relying on the outstanding achievements in the field of incubation equipment and project, EI Electronics applies its development concept and innovative ideas to the research and development, production and service of poultry farm equipment. EI poultry equipment has realized automatic control of poultry house environment, ventilation, heating, humidifying, feeding, drinking, excrement cleaning and other feeding behavior, as well as information management and remote monitoring of poultry farms. And EI has constructed a large-scale intensive, green and ecological modern poultry farm management model.

In recent years, EI Electronics has actively undertaken large-scale poultry farm projects, customized reasonable poultry farm configuration scheme according to customer needs, designed and built integrated steel poultry farms, provided high-end breeding equipment and installation and commissioning services, and built EI AIO field information management system based on the Internet of Things framework.





STEEL STRUCTURE POULTRY HOUSE



Cage-breeding house



Floor-breeding single-decker house



Floor-breeding double-decker house

The design and construction of steel structure poultry house in EI poultry farming project can meet the needs of customer turnkey project.

PRODUCT ADVANTAGE

- Single-span steel structure, span from 12 meters to 20 meters or wider.
- Suitable for flood-breeding, cage-breeding and double-deck poultry houses.
- Design according to wind load and snow load.
- High strength steel is used in the main structure of steel structure, such as truss, column, purlin, etc.
- Wall thermal resistance R12 (2.11 K. m²/W) and roof thermal resistance R18 (3.17 K. m²/W).
- All of them are connected by standard parts. Installation is time-saving and efficient.
- The main structure uses steel economically, 10-11 kg/m².

FEED SUPPLYING SYSTEM

Feed supplying system is the right hand of poultry breeding system, it is necessary for the large-scale and modernized poultry farms.

The feed supplying system is composed of silo, feed pipe, auger, feed outlet, drive mechanism, electronic weighing system, etc.

Feeds in the silo is carried to the hopper in the poultry house by feed supply device, the detection device can be installed at the electric motor and feed dropping bottom, so the system supplies feed automatically. The electronic weighting module transmits the weight information to the controller with decoding processing of EI-BQ/L silo transmitter and display the feed weight.





SILO

Silos are processed by hot dip galvanized steel sheet, top and bottom are tapered, and the design of lower cone can make the feed smoothly flow into the conveying system. Silo around is sealed, to ensure that the feed will not become damp. Weighing module is installed at the bottom of the silo, used for silo electronic weighing.



PARAMETERS

- Capacity: 1.5T~32T
- Material: hot dip galvanized steel sheet, 250g/m²
- Lifespan: 20 years~25 years



24-series silo



30-series silo



36-series silo

The capacity of EI series silo is various, and layer is designed from 1 floor to 4 floors

No.	Diameter(mm)	Model	Legs	(mm) Height(mm)	Layer	Capacity(T)
1	2400	EI-240401	4	4.2	1	5.3
2		EI-240402	4	5.1	2	8.0
3		EI-240403	4	6.0	3	10.6
4	3000	EI-300602	6	5.1	2	14.2
5		EI-300603	6	6.0	3	18.4
7	3600	EI-360602	6	5.1	2	20
8		EI-360603	6	6.0	3	26
9		EI-360604	6	6.9	4	32

PRODUCT ADVANTAGE

- Middle of the barrel is 13mm deep corrugated steel, ensure the precision of installation and leakproofness of assembling.
- Top tilter is 30° inclines; achieve high intensity of roof and large capacity of feed. Bottom is 60° inclines, designed for transmit of Slippery gains and dried feed.
- Bottom through connection is designed to be 15° inclined, in different feed supplying cases, which can connect to be 0° or 30° supplying connector with through connection plate.
- Equip electronic weighing system according to customer request. One silo equips one high precision weighing system.
- Multifunction Compatibility: electronic weighing system can display feed level precisely, with the support of EI feed supplying system, achieve major-minor silo control, precise feed supplying, long-distance supplying, that reduce labor intensity, release the feed car entrance for better epidemic prevention.



Double outlet silo base



Top of silo



FEED SUPPLYING SYSTEM



1 silo for 1 poultry house



1 silo for 2 poultry houses



Outside feed supply

FEEDING SILO

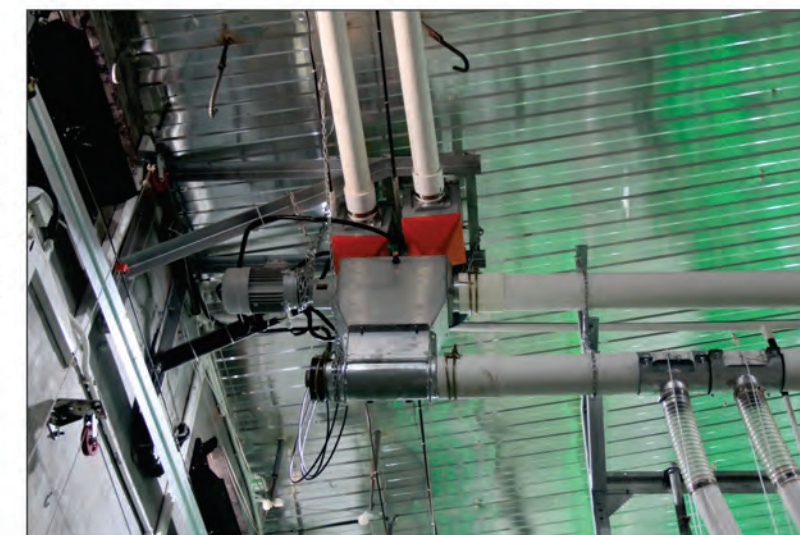
Feeding silo for single poultry house

Feeding silo for double poultry house

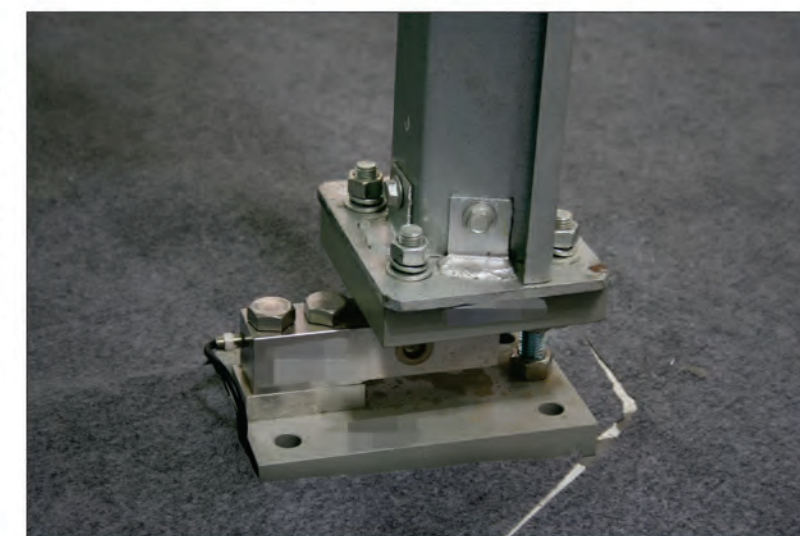
Outside feed supply

PRODUCT ADVANTAGE

- Easy to operate: the user only needs to pour feed into the silo (or feed bin) outside the poultry house.
- Even feed distribution: adjustable feed drop speed ensures consistent amount of feed at each feed outlet.
- Epidemic prevention: separation of silo from the house reduces artificial operation contamination.
- Adequate feed output: facilitates simultaneous feeding of all birds in one house.
- Feed type: pellet, powder
- Safe and reliable: feed supplying system is 4 protection sets to ensure safe and reliable operation of the system. They are silo feed lacking protection, hopper feed full protection, feed spilling protection and motor overload protection.



Feed drive mechanism



Weighing module



Weighing module control box



HEATING SYSTEM

The heating systems of poultry houses are commonly divided in several ways: air energy pump, gas heater, boiler water heating and radiation heating, etc.

AIR ENERGY PUMP HEATING

Air-energy heat pump technology is an energy-saving and environment-friendly heating technology based on the principle of reverse carnot cycle. Low-temperature heat source is obtained through natural energy (air heat storage). After high-efficiency heat collection, it becomes a high-temperature heat source for heating. The whole system has a high heat collection efficiency. This is an energy-saving and environmentally friendly form of energy utilization, which greatly improves energy efficiency and reduces greenhouse gas emissions. Air energy pump technology can be used in cage-breeding house or floor-breeding house heating.



PRODUCT ADVANTAGE

- Energy saving: high energy efficiency ratio, it is the most energy-saving products compared with coal-fired, oil-fired and gas-fired boilers;
- Cost-saving: heat pump units are installed on the roof or outdoor, no need to build a boiler room, save land investment, no annual inspection and operation surcharge, no need for fuel transportation and storage, no need for dedicated duty, no need for complex maintenance and repair, short investment recovery period;
- Safety: heat pump units have no fuel leakage, fire, explosion and other potential safety hazards;
- Environmental: heat pump unit does not pollute the environment, belongs to high-efficiency energy-saving environmental protection products;
- Simple: automatic intelligent control, modular design, simple installation and easy use.





GAS HEATER

Gas heater has the characteristics of environmental protection, energy saving, safety and full automatic operation. Based on the support of relevant national policies, gas-fired boilers will be more widely used. Gas heater can be used in cage-breeding house or floor-breeding house.



□□□□□□□□
Indoor gas heater



Outdoor gas heater

PRODUCT ADVANTAGE

- Heater power range from 17.6 ~ 73 KW.
- Fuel uses LPG, NG and butane.
- There are several types of ignition, such as high temperature ignition, spark ignition and direct ignition.
- The case can be painted galvanized steel or stainless steel.

RADIANT HEATING BURNER



PRODUCT ADVANTAGE

- The heat output range is 23.4~44.0KW.
- An intelligent flame monitoring system is installed inside to ensure the reliability of equipment combustion.
- Available in single or double tube models.
- The fuel has two types of natural gas or liquid propane gas.
- Electronic ignition

HEATING OF WATER CIRCULATION BOILER

The water circulating boiler in poultry house mainly uses the hot water radiator to heat the poultry house. The indoor temperature is raised to 1-40 °C, and the combustion stops automatically after reaching the temperature. Water circulating boiler can be used in cage breeding house or floor-breeding house for heating.

The main pipe of the heating boiler is leaded into the house, and the radiators are arranged on both sides of the house, and are evenly arranged. The radiators can be partitioned according to actual needs. When the real time temperature drops to the heating temperature, the radiator in this area starts to heat automatically; when the real-time temperature returns to the target temperature, the radiator in this area stops heating automatically.

Heating method is divided into the direct return and reverse return. Direct return heating indicates the water supply pipe and the return pipe have the same water flow direction, which can avoid the overheating or non-heating. The heating effect is ideal, but with the length of the return pipe increasing, and the investment cost goes up. Reverse return heating is totally different. Water direction of the supply water and return water is opposite. Although the length of the return pipe is short, easy to construct, and the investment cost is small, it also causes the problem that the head of pipe is overheated and the end of pipe is not hot, and the heating effect is not satisfactory.





ENGINEERING CASE

